

Project Data Page:





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Ammonia Scrubber

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Process Cond StripperOpR2.pvdb

Date of Analysis : Apr 18,2026 11:31am

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2750.000 mm ID	
Vessel Design Length, Tangent to Tangent	19820.00 mm
Distance of Bottom Tangent above Grade	6400.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	6100.00 mm
Internal Design Temperature	430 °C
Internal Design Pressure	5929.490 kPa
External Design Temperature	200 °C
External Design Pressure	103.421 kPa
Maximum Allowable Working Pressure	6068 kPa
External Max. Allowable Working Pressure	1408 kPa
Shop Test Pressure	8054.110 kPa
Field Test Pressure	7760.377 kPa
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	39.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	+IT
Head	SA-387 11	2	...	K11789	+IT
Skirt	SA-516 70	...	...	K02700	
Skirt	SA-387 11	2	...	K11789	
Nozzle	SA-336 F11	3	...	K11572	+IT
Nozzle	SA-387 11	2	...	K11789	
Nozzle	SA-335 P11	...	...	K11597	
Re-Pad	SA-387 11	2	...	K11789	
Nozzle Flg	SA-182 F11	2	...	K11572	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
K-3	...	...	...	1.6000	N/A	No
K-2	...	...	...	1.6000	N/A	No
K-1	...	...	...	1.6000	N/A	No
H-1	5961.171	103.42	6091.000	1.0000	No	No
S-1	5955.605	103.42	6119.000	1.0000	N/A	No
S-2	5932.962	103.42	6141.000	1.0000	N/A	No
S-3	5931.830	103.42	6142.000	1.0000	N/A	No
S-4	5931.830	103.42	6142.000	1.0000	N/A	No
S-5	5931.830	103.42	6142.000	1.0000	N/A	No
S-6	5931.830	103.42	6142.000	1.0000	N/A	No



S-7	5931.830	103.42	6142.000	1.0000	N/A	No
H-2	5931.830	103.42	6068.000	1.0000	No	No

Liquid Level: 3887.50 mm Dens.: 769.647 kg/m³ Sp. Gr.: 0.770

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-3600.0	2500.0	20.0	20.0	...	...	0.70	0.70
Skirt	-600.0	3000.0	18.0	18.0	...	...	0.70	0.70
Skirt	0.0	600.0	18.0	18.0	...	...	0.70	0.50
Ellipse	50.0	50.0	68.0	58.5	57.0	8.5	1.00	1.00
Cylinder	3050.0	3000.0	60.0	60.0	58.1	22.4	1.00	1.00
Cylinder	6050.0	3000.0	60.0	60.0	57.9	22.4	1.00	1.00
Cylinder	9050.0	3000.0	60.0	60.0	57.9	22.4	1.00	1.00
Cylinder	12050.0	3000.0	60.0	60.0	57.9	22.4	1.00	1.00
Cylinder	15050.0	3000.0	60.0	60.0	57.9	22.4	1.00	1.00
Cylinder	18050.0	3000.0	60.0	60.0	57.9	22.4	1.00	1.00
Cylinder	19770.0	1720.0	60.0	60.0	57.9	22.4	1.00	1.00
Ellipse	19820.0	50.0	68.0	58.0	56.7	8.5	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
D-10"	WRC-107/537	0.894	n/a	Passed
E-8"	WRC-297	0.987	Shell	Passed
B-14"	WRC-297	0.984	Shell	Passed
A-10"	WRC-297	0.984	Shell	Passed
C-14"	WRC-107/537	0.897	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	1250	6666.03	32296.1	95724.6	608251
20	30	4000	5991.29	32217.2	79903	527610
30	40	5800	5550.88	32022.4	62589.7	431250
40	50	6125	5462.8	31968.6	59285.6	412053
50	60	7650	5335.18	31490.1	59015.6	410466
60	70	10650	4617.62	29120	44086.4	319551
70	80	13650	3848.82	26810.8	31386.8	235655
80	90	16650	3062.1	23920.1	21020.4	159558
90	100	19650	2553.68	18268.8	12596.7	96274.9
100	110	22650	1798.98	13849.3	6067.69	48097.7
110	120	25010	1031.53	8426.79	1821.92	14683.6
120	130	25895	740.041	5887.19	298.367	2373.57

Abs Max of the all of the Stress Ratio's : 0.7591

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	44.0000	mm
Inside Diameter of Basing	2430.0000	mm
Outside Diameter of Basing	3030.0000	mm
Nominal Diameter of Bolts	64.0000	mm
Diameter of Bolt Circle	2837.9998	mm
Number of Bolts	20	

Thickness of Gusset Plates	20.0000	mm
Average Width of Gusset Plates	182.0000	mm
Height of Gussets	312.0000	mm



Distance between Gussets	120.0000	mm
Thickness of Top Plate or Ring	50.0000	mm
Radial Width of the Top Plate	192.0000	mm

Basering Sketch for Cylindrical Skirts:

Basering Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	10274.	kgf
Total Earthquake Shear on Support	35904.	kgf
Wind Moment on Support	236044.	kgf-m.
Earthquake Moment on Support	748571.	kgf-m.
Maximum Initial Bolt Load	41818.	kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	132797.1	kg
Shop Test - Fabricated + water (full)	261001.7	kg
Shipping - Fab. + rem. intls.+ shipping app.	167615.0	kg
Erected - Fab. + rem. intls.+ insul. (etc)	173275.0	kg
Empty - Fab. + intls. + details + wghts.	181418.2	kg
Operating - Empty + operating liquid (no ca)	222136.3	kg
Field Test - Empty weight + test medium (full)	295819.6	kg

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Nozzle Calculation Summary (kPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
D-10"	6091	OK	...	OK	8.10	OK	Passed
MH-3-24"	6093	OK	...	...		OK	Passed
E-8"	6115	OK	...	OK	7.16	OK	Passed
UC-2-14"	6129	OK	...	...		OK	Passed
B-14"	6129	OK	...	OK	8.33	OK	Passed
MH-2-24"	6093	OK	...	...		OK	Passed
UC-1-14"	6129	OK	...	...		OK	Passed
A-10"	6075	OK	...	OK	8.17	OK	Passed
MH-1-24"	6093	OK	...	...		OK	Passed
C-14"	6068	OK	...	OK	8.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6068 Nozzle : C-14"
Minimum MAWP Shells/Flanges : 6068 Element : H-2
Minimum MAPnc Shells/Flanges : 6216 Element : H-2

Computed Vessel M.A.W.P. : 6068 kPa

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
40	D-10"	0.000	0.000	493.293
50	MH-3-24"	1550.000	180.000	1099.200
50	E-8"	550.000	0.000	357.607
60	UC-2-14"	5250.000	0.000	639.200
60	B-14"	3650.000	0.000	639.200
80	MH-2-24"	10550.000	0.000	1099.200
90	UC-1-14"	12550.001	0.000	639.200
100	A-10"	16900.004	0.000	446.087
110	MH-1-24"	18910.004	0.000	1099.200
120	C-14"	0.000	0.000	647.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Thk mm	Cut Length mm	Flg Class
E-8"	8.000 in	160	WNF	8.625	23.012	819.08	64.00	564.37	600
D-10"	10.000 in	80	WNF	10.750	15.088	...	...	312.20	600
A-10"	10.000 in	160	WNF	10.750	28.575	593.05	62.00	566.79	600
UC-2-14"	14.000 in Actual		WNF	14.000	18.000	655.60	58.00	571.54	600
B-14"	14.000 in Actual		WNF	14.000	18.000	655.60	58.00	571.54	600
UC-1-14"	14.000 in Actual		WNF	14.000	18.000	655.60	58.00	571.54	600
C-14"	14.000 in Actual		WNF	14.000	16.000	655.60	58.00	564.28	900
MH-3-24"	24.000 in Actual		WNF	24.000	30.000	1049.60	58.00	594.21	600
MH-2-24"	24.000 in Actual		WNF	24.000	30.000	1049.60	58.00	594.21	600
MH-1-24"	24.000 in Actual		WNF	24.000	30.000	1049.60	58.00	594.21	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
E-8"	SA-335 P11	60.000	20.000	60.000	60.000	...
D-10"	SA-336 F11	58.500	10.000	...	...	...
A-10"	SA-335 P11	60.000	20.000	46.000	60.000	...
UC-2-14"	SA-387 11	58.000	10.000	44.000	58.000	...
B-14"	SA-387 11	58.000	10.000	44.000	58.000	...
UC-1-14"	SA-387 11	58.000	10.000	44.000	58.000	...
C-14"	SA-387 11	58.000	10.000	44.000	58.000	...
MH-3-24"	SA-387 11	58.000	10.000	44.000	58.000	...
MH-2-24"	SA-387 11	58.000	10.000	44.000	58.000	...
MH-1-24"	SA-387 11	58.000	10.000	44.000	58.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
E-8"	550.000	0.0	500.00	0.00	S-1
D-10"	...	0.0	250.00	0.00	H-1
A-10"	16900.002	0.0	500.00	0.00	S-6
UC-2-14"	5250.000	0.0	500.00	0.00	S-2
B-14"	3650.000	0.0	500.00	0.00	S-2
UC-1-14"	12550.001	0.0	500.00	0.00	S-5
C-14"	...	0.0	500.00	0.00	H-2
MH-3-24"	1550.000	180.0	500.00	0.00	S-1
MH-2-24"	10550.000	0.0	500.00	0.00	S-4
MH-1-24"	18910.004	0.0	500.00	0.00	S-7



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 68.0mm THK X 2750.0mm ID X 50.0mm	SA-387 11
6	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 2000.0mm	...
1	INSULATION: 2500mm X 50mm THK	
1	LINING: 2500.000mm X 50.000mm THK	...
1	INSULATION: 3000mm X 50mm THK	
1	LINING: 3000.000mm X 50.000mm THK	...
1	INSULATION: 600mm X 50mm THK	
1	LINING: 600.000mm X 50.000mm THK	...
2	INSULATION: 737mm X 150mm THK	hot
2	LINING: 737.500mm X 3.000mm THK	...
6	LADDER: 4000.0mm	...
6	INSULATION: 3000mm X 150mm THK	hot
6	LINING: 3000.000mm X 3.000mm THK	...
1	PACKING: 1175.000mm	...
2	PACKING: 3000.000mm	...
1	PACKING: 325.001mm	...
1	PACKING: 845.000mm	...
1	PACKING: 655.001mm	...
1	INSULATION: 1720mm X 150mm THK	hot
1	LINING: 1720.000mm X 3.000mm THK	...
1	PLATFORM: 1000mm X 6999mm WIDE	
3	CLASS 600 GR 1.9, 24.0" BLIND FLANGE(S)	SA-182 F11
2	CLASS 600 GR 1.9, 14.0" BLIND FLANGE(S)	SA-182 F11
1	BASERING: 3030.0mm OD X 2430.0 ID X 44.0mm	SA-516 70
20	BASERING BOLTS: 64.0mm DIA	SA-36
20	BASERING BOLT NUTS: 64.0mm DIA	...
1	BASERING TOP RING: 3030.0mm OD X 2646.0 ID X 50.0mm	...
40	BASERING GUSSET PLATES: 182.0mm X 182.0mm X 312.0mm X 20.0mm	...
2	CYLINDRICAL SKIRT SEGMENT 20.0mm THK X 3001.8mm X 2500.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 20.0mm THK X 2183.2mm X 2500.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3006.5mm X 3000.0mm	SA-516 70



1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2186.5mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2892.8mm X 600.0mm	SA-387 11
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2413.9mm X 600.0mm	SA-387 11
9	CYLINDRICAL SEGMENT 60.0mm THK X 3048.0mm X 3000.0mm	SA-387 11
3	CYLINDRICAL SEGMENT 60.0mm THK X 2543.3mm X 3000.0mm	SA-387 11
3	CYLINDRICAL SEGMENT 60.0mm THK X 5591.3mm X 3000.0mm	SA-387 11
3	CYLINDRICAL SEGMENT 60.0mm THK X 0.0mm X 3000.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 60.0mm THK X 5591.3mm X 1720.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 60.0mm THK X 3048.0mm X 1720.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 60.0mm THK X 0.0mm X 1720.0mm	SA-387 11
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
H-1	[10]	!	-12	-13		0.989	58.500	1.00	Yes
H-1	[7]	!	-12	-19		0.868	68.000	1.00	Yes
S-1	[8]	!	-12	-13		0.985	60.000	1.00	Yes
S-2	[8]	!	-12	-13		0.981	60.000	1.00	Yes
S-3	[8]	!	-12	-13		0.981	60.000	1.00	Yes
S-4	[8]	!	-12	-13		0.981	60.000	1.00	Yes
S-5	[8]	!	-12	-13		0.981	60.000	1.00	Yes
S-6	[8]	!	-12	-13		0.981	60.000	1.00	Yes
S-7	[8]	!	-12	-13		0.981	60.000	1.00	Yes
H-2	[10]	!	-12	-12		0.993	58.000	1.00	Yes
H-2	[7]	!	-12	-20		0.864	68.000	1.00	Yes
D-10"	[1]	!	-12	-18		0.891	58.500	1.00	Yes
Nozzle Flg	[5]	B	-18	-41					
MH-3-24"	[1]	A	39	39		0.983	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-41					
E-8"	[1]	A	40	39		0.984	60.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-41					
UC-2-14"	[1]	A	39	38		0.981	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-41					
B-14"	[1]	A	39	38		0.981	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-41					
MH-2-24"	[1]	A	39	38		0.981	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-41					
UC-1-14"	[1]	A	39	38		0.981	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-41					
A-10"	[1]	A	40	39		0.981	60.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-41					
MH-1-24"	[1]	A	39	38		0.981	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-41					
C-14"	[1]	A	39	34		0.894	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-48					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No
Warmest MDMT:			40	39					

Required Minimum Design Metal Temperature -12.0 °C
Warmest Computed Minimum Design Metal Temperature 39.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld



[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 30 50 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (QEUP):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	kPa
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	16.0183	kg/m ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	0.4470	m/sec
Tray Weight	lbs./sq.ft.	47.8790	N/m ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Ammonia Scrubber

Design Internal Pressure (for Hydrotest)	5929.5	kPa
Design Internal Temperature	430.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	Amine	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes



Complete Listing of Vessel Elements and Details:

Element#	1/12	
Element From Node	10	
Element To Node	20	
Element Type	Skirt Support	
Description	K-3	
Distance "FROM" to "TO"	2500	mm
Skirt Outside Diameter	2646	mm
Diameter of Skirt at Base	2646	mm
Skirt Thickness	20	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2473.7	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Platform	
Detail ID	PL-6	
Dist. from "FROM" Node / Offset dist	2100	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	23128	cm ²
Platform Weight	982.95	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	175	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	2000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm
Element From Node	10	
Detail Type	Insulation	
Detail ID	FR-3	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2500	mm
Thickness of Insulation	50	mm
Density	800	kg/m ³
Element From Node	10	
Detail Type	Lining	
Detail ID	FR-3.1	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	2500	mm
Thickness of Lining	50	mm
Density	800	kg/m ³
Element From Node	10	



Detail Type	Weight
Detail ID	Taling Lug
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	1000 kgf
Offset from Element Centerline	0 mm

Element#	2/12
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	K-2
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	2646 mm
Diameter of Skirt at Base	2646 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element From Node	20
Detail Type	Insulation
Detail ID	FR-2
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	50 mm
Density	800 kg/m ³

Element From Node	20
Detail Type	Lining
Detail ID	FR-2.1
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	50 mm
Density	800 kg/m ³

Element From Node	20
Detail Type	Weight
Detail ID	Skirt MW
Dist. from "FROM" Node / Offset dist	0.0002896 mm
Miscellaneous Weight	500 kgf
Offset from Element Centerline	0 mm

Element#	3/12
Element From Node	30
Element To Node	40
Element Type	Skirt Support
Description	K-1
Distance "FROM" to "TO"	600 mm
Skirt Outside Diameter	2646 mm
Diameter of Skirt at Base	2646 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	200 °C
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11



Allowable Stress, Ambient	1509.1	kgf/cm ²
Allowable Stress, Operating	1509.1	kgf/cm ²
Allowable Stress, Hydrotest	2844.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	16.002	mm
Yield Stress, Operating	2671.6	kgf/cm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Insulation	
Detail ID	FR-1	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	600	mm
Thickness of Insulation	50	mm
Density	800	kg/m ³

Element From Node	30	
Detail Type	Lining	
Detail ID	FR-1.1	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	600	mm
Thickness of Lining	50	mm
Density	800	kg/m ³

Element#	4/12	
Element From Node	40	
Element To Node	50	
Element Type	Elliptical	
Description	H-1	
Distance "FROM" to "TO"	50	mm
Inside Diameter	2750	mm
Element Thickness	58.5	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	68	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5929.5	kPa
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	kPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 10	
Dist. from "FROM" Node / Offset dist	-687.5	mm
Height/Length of Liquid	737.5	mm
Liquid Density	769.65	kg/m ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-687.5	mm
Height/Length of Insulation	737.5	mm
Thickness of Insulation	150	mm
Density	352	kg/m ³



Element From Node	40
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	-687.5 mm
Height/Length of Lining	737.5 mm
Thickness of Lining	3 mm
Density	8027 kg/m ³

Element From Node	40
Detail Type	Nozzle
Detail ID	D-10"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	10 in.
Nozzle Schedule	80
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	466.92 kgf
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-336 F11 +IT

Element From Node	40
Detail Type	Weight
Detail ID	PPD
Dist. from "FROM" Node / Offset dist	-685 mm
Miscellaneous Weight	196 kgf
Offset from Element Centerline	2840 mm

Element#	5/12
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	S-1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	2750 mm
Element Thickness	60 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	60 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5929.5 kPa
Design Temperature Internal Pressure	430 °C
Design External Pressure	103.42 kPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Platform
Detail ID	PL-1
Dist. from "FROM" Node / Offset dist	0.48768E-04 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	360.0
Platform Wind Area	30000 cm ²
Platform Weight	3931.3 kgf
Platform Railing Weight	0 kgf/mm
Platform Grating Weight	1668 N/m ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	175 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm



Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	769.65	kg/m ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	150	mm
Density	352	kg/m ³

Element From Node	50	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	MH-3-24"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1453.4	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-387 11	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	E-8"	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	8	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	396.7	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-335 P11	

Element From Node	50	
Detail Type	Weight	
Detail ID	PPE	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	1349	kgf
Offset from Element Centerline	2815	mm

Element#	6/12	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	S-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	2750	mm
Element Thickness	60	mm



Internal Corrosion Allowance	1	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5929.5	kPa
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	kPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Platform	
Detail ID	PL-2	
Dist. from "FROM" Node / Offset dist	1610	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	30000	cm ²
Platform Weight	2039.6	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	175	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	60	
Detail Type	Packing	
Detail ID	Bed#2	
Dist. from "FROM" Node / Offset dist	1825	mm
Height of Packed Section	1175	mm
Density	153	kg/m ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	0.76999998	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	150	mm
Liquid Density	769.65	kg/m ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	150	mm
Density	352	kg/m ³

Element From Node	60	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³

Element From Node	60	
Detail Type	Nozzle	
Detail ID	UC-2-14"	
Dist. from "FROM" Node / Offset dist	2200	mm

Nozzle Diameter	14	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	479.06	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-387 11	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	B-14"	
Dist. from "FROM" Node / Offset dist	600	mm
Nozzle Diameter	14	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	324.33	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-387 11	

Element From Node	60	
Detail Type	Weight	
Detail ID	DLBS-1	
Dist. from "FROM" Node / Offset dist	1825	mm
Miscellaneous Weight	630	kgf
Offset from Element Centerline	0	mm

Element From Node	60	
Detail Type	Weight	
Detail ID	PPB	
Dist. from "FROM" Node / Offset dist	600	mm
Miscellaneous Weight	2788	kgf
Offset from Element Centerline	2891	mm

Element#	7/12	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	S-3	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	2750	mm
Element Thickness	60	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5929.5	kPa
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	kPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Platform	
Detail ID	PL-3	
Dist. from "FROM" Node / Offset dist	2610	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	2039.6	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²



Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	175	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	70	
Detail Type	Packing	
Detail ID	Bed#2	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	3000	mm
Density	153	kg/m ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	0.76999998	

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	150	mm
Density	352	kg/m ³

Element From Node	70	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³

Element#	8/12	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	S-4	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	2750	mm
Element Thickness	60	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5929.5	kPa
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	kPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Packing	
Detail ID	Bed#2	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	325	mm
Density	153	kg/m ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	0.76999998	

Element From Node	80	
Detail Type	Packing	



Detail ID	Bed#1
Dist. from "FROM" Node / Offset dist	2155 mm
Height of Packed Section	845 mm
Density	153 kg/m ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	0.76999998

Element From Node	80
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	150 mm
Density	352 kg/m ³

Element From Node	80
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	3 mm
Density	8027 kg/m ³

Element From Node	80
Detail Type	Nozzle
Detail ID	MH-2-24"
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	1453.4 kgf
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-387 11

Element From Node	80
Detail Type	Weight
Detail ID	DLBS-2
Dist. from "FROM" Node / Offset dist	2155 mm
Miscellaneous Weight	630 kgf
Offset from Element Centerline	0 mm

Element From Node	80
Detail Type	Weight
Detail ID	DLLD-2
Dist. from "FROM" Node / Offset dist	800 mm
Miscellaneous Weight	800 kgf
Offset from Element Centerline	0 mm

Element From Node	80
Detail Type	Weight
Detail ID	DLBL-2
Dist. from "FROM" Node / Offset dist	325 mm
Miscellaneous Weight	160 kgf
Offset from Element Centerline	0 mm

Element From Node	80
Detail Type	Weight
Detail ID	OPLD-2
Dist. from "FROM" Node / Offset dist	800 mm
Miscellaneous Weight	1100 kgf
Offset from Element Centerline	0 mm

Element From Node	80
Detail Type	Weight
Detail ID	TALD-2
Dist. from "FROM" Node / Offset dist	770 mm
Miscellaneous Weight	100 kgf



Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	small Noz	
Dist. from "FROM" Node / Offset dist	950	mm
Miscellaneous Weight	160	kgf
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	Cleat Weight	
Dist. from "FROM" Node / Offset dist	950	mm
Miscellaneous Weight	4252	kgf
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	Margin Weight	
Dist. from "FROM" Node / Offset dist	950	mm
Miscellaneous Weight	10000	kgf
Offset from Element Centerline	0	mm

Element#	9/12	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	S-5	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	2750	mm
Element Thickness	60	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5929.5	kPa
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	kPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Platform	
Detail ID	PL-4	
Dist. from "FROM" Node / Offset dist	610	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	30000	cm ²
Platform Weight	2039.6	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	175	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	90	
Detail Type	Packing	
Detail ID	Bed#1	



Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	3000	mm
Density	153	kg/m ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	0.76999998	

Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	150	mm
Density	352	kg/m ³

Element From Node	90	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³

Element From Node	90	
Detail Type	Nozzle	
Detail ID	UC-1-14"	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	14	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	479.06	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-387 11	

Element#	10/12	
Element From Node	100	
Element To Node	110	
Element Type	Cylinder	
Description	S-6	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	2750	mm
Element Thickness	60	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5929.5	kPa
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	kPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	100	
Detail Type	Platform	
Detail ID	PL-5	
Dist. from "FROM" Node / Offset dist	1610	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	2039.6	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm



Platform Height	1000	mm
Platform Clearance or End Offset	175	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	100	
Detail Type	Packing	
Detail ID	Bed#1	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	655	mm
Density	153	kg/m ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	0.76999998	

Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	150	mm
Density	352	kg/m ³

Element From Node	100	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³

Element From Node	100	
Detail Type	Nozzle	
Detail ID	A-10"	
Dist. from "FROM" Node / Offset dist	1850	mm
Nozzle Diameter	10	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	311.06	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-335 P11	

Element From Node	100	
Detail Type	Weight	
Detail ID	DLLD-1	
Dist. from "FROM" Node / Offset dist	1110	mm
Miscellaneous Weight	800	kgf
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	DLBL-1	
Dist. from "FROM" Node / Offset dist	655	mm
Miscellaneous Weight	160	kgf
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	DLPF	
Dist. from "FROM" Node / Offset dist	1850	mm
Miscellaneous Weight	250	kgf
Offset from Element Centerline	0	mm

Element From Node	100	
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Detail Type	Weight
Detail ID	OPLD-1
Dist. from "FROM" Node / Offset dist	1110 mm
Miscellaneous Weight	1100 kgf
Offset from Element Centerline	0 mm

Element From Node	100
Detail Type	Weight
Detail ID	OPFP
Dist. from "FROM" Node / Offset dist	1850 mm
Miscellaneous Weight	140 kgf
Offset from Element Centerline	0 mm

Element From Node	100
Detail Type	Weight
Detail ID	TALD-1
Dist. from "FROM" Node / Offset dist	1200 mm
Miscellaneous Weight	100 kgf
Offset from Element Centerline	0 mm

Element From Node	100
Detail Type	Weight
Detail ID	PPA
Dist. from "FROM" Node / Offset dist	1850 mm
Miscellaneous Weight	3535 kgf
Offset from Element Centerline	2840 mm

Element#	11/12
Element From Node	110
Element To Node	120
Element Type	Cylinder
Description	S-7
Distance "FROM" to "TO"	1720 mm
Inside Diameter	2750 mm
Element Thickness	60 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	60 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5929.5 kPa
Design Temperature Internal Pressure	430 °C
Design External Pressure	103.42 kPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	110
Detail Type	Insulation
Detail ID	Ins: 80
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1720 mm
Thickness of Insulation	150 mm
Density	352 kg/m ³

Element From Node	110
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	1720 mm
Thickness of Lining	3 mm
Density	8027 kg/m ³

Element From Node	110
Detail Type	Nozzle
Detail ID	MH-1-24"



Dist. from "FROM" Node / Offset dist	860	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1453.4	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-387 11	

Element From Node	110	
Detail Type	Weight	
Detail ID	DLPAD	
Dist. from "FROM" Node / Offset dist	1350	mm
Miscellaneous Weight	230	kgf
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	TAPAD	
Dist. from "FROM" Node / Offset dist	1350	mm
Miscellaneous Weight	200	kgf
Offset from Element Centerline	0	mm

Element#	12/12	
Element From Node	120	
Element To Node	130	
Element Type	Elliptical	
Description	H-2	
Distance "FROM" to "TO"	50	mm
Inside Diameter	2750	mm
Element Thickness	58	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	68	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5929.5	kPa
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	kPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	120	
Detail Type	Platform	
Detail ID	TOP PL	
Dist. from "FROM" Node / Offset dist	890	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	70000	cm ²
Platform Weight	8482.6	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	7000	mm
Platform Height	1000	mm
Platform Clearance or End Offset	3500	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	7000	mm

Element From Node	120	
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Detail Type	Insulation
Detail ID	Ins: 90
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	737.5 mm
Thickness of Insulation	150 mm
Density	352 kg/m ³
Element From Node	120
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	737.5 mm
Thickness of Lining	3 mm
Density	8027 kg/m ³
Element From Node	120
Detail Type	Nozzle
Detail ID	C-14"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	14 in.
Nozzle Schedule	None
Nozzle Class	900
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	377.95 kgf
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-387 11
Element From Node	120
Detail Type	Weight
Detail ID	PPC
Dist. from "FROM" Node / Offset dist	687 mm
Miscellaneous Weight	3578 kgf
Offset from Element Centerline	2891 mm
Element From Node	120
Detail Type	Weight
Detail ID	PSV
Dist. from "FROM" Node / Offset dist	642 mm
Miscellaneous Weight	2000 kgf
Offset from Element Centerline	3500 mm
Element From Node	120
Detail Type	Weight
Detail ID	Lifting Lug
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	1500 kgf
Offset from Element Centerline	0 mm
Element From Node	120
Detail Type	Weight
Detail ID	Pipe Davit
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	600 kgf
Offset from Element Centerline	1400 mm
Element From Node	120
Detail Type	For./Mom.
Detail ID	CFxFZ
Dist. from "FROM" Node / Offset dist	685 mm
Force in X Direction	2550.9 kgf
Force in Y Direction	-3532.1 kgf
Force in Z Direction	2550.9 kgf
Moment about X Axis	2746.7 kgf-m.
Moment about Y Axis	0 kgf-m.
Moment about Z Axis	2746.7 kgf-m.
Force/Moment Combination Method	SRSS



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	K-3	...	2500	...	2500
	K-2	...	5500	...	3000
	K-1	...	6100	...	600
	H-1	...	6150	...	50
	S-1	...	9150	...	3000
	S-2	...	12150	...	3000
	S-3	...	15150	...	3000
	S-4	...	18150	...	3000
	S-5	...	21150	...	3000
	S-6	...	24150	...	3000
	S-7	...	25870	...	1720
	H-2	...	25920	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
	K-3	...	20	1.6	2646	...	1
	K-2	...	18	1.6	2646	...	2
	K-1	...	18	1.6	2646	...	3
	H-1	5961.2	68	1	2750	1499	4
	S-1	5955.6	60	1	2750	1499	5
	S-2	5933	60	1	2750	1499	6
	S-3	5931.8	60	1	2750	1499	7
	S-4	5931.8	60	1	2750	1499	8
	S-5	5931.8	60	1	2750	1499	9
	S-6	5931.8	60	1	2750	1499	10
	S-7	5931.8	60	1	2750	1499	11
	H-2	5931.8	68	1	2750	1499	12

Element Required Thickness and MAWP :

From	To	Design Pressure kPa	M.A.W.P. Corroded kPa	M.A.P. New & Cold kPa	Minimum Thickness mm	Required Thickness mm	Element#
	K-3	...	No Calc	No Calc	20	No Calc	1
	K-2	...	No Calc	No Calc	18	No Calc	2
	K-1	...	No Calc	No Calc	18	No Calc	3
	H-1	5929.49	6091	6270	58.5	56.9742	4
	S-1	5929.49	6119	6293	60	58.1381	5
	S-2	5929.49	6141	6293	60	57.9155	6
	S-3	5929.49	6142	6293	60	57.9044	7
	S-4	5929.49	6142	6293	60	57.9044	8
	S-5	5929.49	6142	6293	60	57.9044	9
	S-6	5929.49	6142	6293	60	57.9044	10
	S-7	5929.49	6142	6293	60	57.9044	11
	H-2	5929.49	6068	6216	58	56.6976	12

Minimum 6068 6216

MAWP: 6068 kPa, limited by: H-2.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: H-1

Elliptical Head Design Information:					
Design Pressure	P	5961.171 kPa	Int. Design Temperature		430 °C
Material		SA-387 11	Ext. Design Temperature		200 °C
Length		50.000 mm	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1499.0 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm ²
Inside Diameter	Di	2750.000 mm	Actual Stress		1459.4 kgf/cm ²
Spec. Min. Thk t _e , or s _a		58.5000 mm	Spec. Nominal Thk.		68.000 mm
Surface Area		94818.4 cm ²	SG of Contents		0.770
Element Volume		3019284.5 cm ³	Weight of Contents		2309.1 kgf
Empty Weight		5242.3 kgf	Operating Weight		7551.4 kgf
			Comp. All. Ope. Stress		823.8 kgf/cm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 34.375 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.188 mm

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (5961 \cdot 2752 \cdot 1) / (2 \cdot 1499 \cdot 1 - 0.2 \cdot 5961)$$

$$= 55.9742 + 1.0000 = 56.9742 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 31.681 kPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1499 \cdot 1 \cdot 57.5) / (1 \cdot 2752 + 0.2 \cdot 57.5)$$

$$= 6123 - 31.7 = 6091 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1509 \cdot 1 \cdot 58.5) / (1 \cdot 2750 + 0.2 \cdot 58.5)$$

$$= 6270 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (5961 \cdot (1 \cdot 2752 + 0.2 \cdot 57.5)) / (2 \cdot 1 \cdot 57.5)$$

$$= 1459.365 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (5961 \cdot 1376) / (1499 \cdot 1 - 0.6 \cdot 5961) + 1$$

$$= 58.193 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 26.492 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1499 \cdot 1 \cdot 67) / (1376 + 0.6 \cdot 67)$$

$$= 6954 - 26.5 = 6928 \text{ kPa}$$



Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (2752 / (2 \cdot 689))^2) / 6$
 $= 0.999032$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 10.169 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 34.375 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.188 mm

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: H-1

Elliptical Head From 40 to 50 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
57.500	2867.00	49.86	0.0027855	1002.40

$$\text{MAEP} = B / (K0 \cdot Do / t) = 1002 / (0.9 \cdot 49.9) = 2191 \text{ kPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
7.544	2867.00	380.05	0.0003655	360.75

$$\text{MAEP} = B / (K0 \cdot Do / t) = 361 / (0.9 \cdot 380) = 103 \text{ kPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: S-1

Cylindrical Shell Design Information:				
Design Pressure	P	5955.605 kPa	Int. Design Temperature	430 °C
Material		SA-387 11	Ext. Design Temperature	200 °C
Length		3000.000 mm	External Pressure Chart	CS-3
			UNS Number	K11789
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope) S	1499.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1509.1 kgf/cm²
Inside Diameter	Di	2750.000 mm	Actual Stress	1452.8 kgf/cm²
Spec. Min. Thk t _e , or sa		60.0000 mm	Spec. Nominal Thk.	60.000 mm
Surface Area		270491.2 cm²	SG of Contents	0.770
Element Volume		17818726.0 cm³	Weight of Contents	13654.5 kgf
Empty Weight		12315.5 kgf	Operating Weight	25970.0 kgf
			Comp. All. Ope. Stress	827.0 kgf/cm²

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (5956 \cdot 1376) / (1499 \cdot 1 - 0.6 \cdot 5956)$
 $= 57.1381 + 1.0000 = 58.1381 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
[Less Operating Hydrostatic Head Pressure of 26.115 kPa](#)
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1499 \cdot 1 \cdot 59) / (1376 + 0.6 \cdot 59)$
 $= 6145 - 26.1 = 6119 \text{ kPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1509 \cdot 1 \cdot 60) / (1375 + 0.6 \cdot 60)$
 $= 6293 \text{ kPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$
 $= (5956 \cdot (1376 + 0.6 \cdot 59)) / (1 \cdot 59)$
 $= 1452.826 \text{ kgf/cm}^2$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: S-1



Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*524)/(3*48.6) = 1408 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*106)/(3*134) = 103 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463)/(3*48.6) = 1245 \text{ kPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: S-2

Cylindrical Shell Design Information:					
Design Pressure	P	5932.962 kPa	Int. Design Temperature	430 °C	
Material		SA-387 11	Ext. Design Temperature	200 °C	
Length		3000.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1499.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	2750.000 mm	Actual Stress	1447.3 kgf/cm²	
Spec. Min. Thk t _e , or sa		60.0000 mm	Spec. Nominal Thk.	60.000 mm	
Surface Area		270491.2 cm²	SG of Contents	0.770	
Element Volume		17818726.0 cm³	Weight of Contents	682.7 kgf	
Empty Weight		12315.5 kgf	Operating Weight	12998.2 kgf	
			Comp. All. Ope. Stress	827.0 kgf/cm²	

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5933 \cdot 1376) / (1499 \cdot 1 - 0.6 \cdot 5933)$$

$$= 56.9155 + 1.0000 = 57.9155 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

[Less Operating Hydrostatic Head Pressure of 3.472 kPa](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1499 \cdot 1 \cdot 59) / (1376 + 0.6 \cdot 59)$$

$$= 6145 - 3.47 = 6141 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1509 \cdot 1 \cdot 60) / (1375 + 0.6 \cdot 60)$$

$$= 6293 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (5933 \cdot (1376 + 0.6 \cdot 59)) / (1 \cdot 59)$$

$$= 1447.302 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: S-2



Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*524)/(3*48.6) = 1408 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*106)/(3*134) = 103 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463)/(3*48.6) = 1245 \text{ kPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: S-3

Cylindrical Shell Design Information:					
Design Pressure	P	5931.830 kPa	Int. Design Temperature	430 °C	
Material		SA-387 11	Ext. Design Temperature	200 °C	
Length		3000.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1499.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	2750.000 mm	Actual Stress	1447.0 kgf/cm²	
Spec. Min. Thk t _e , or sa		60.0000 mm	Spec. Nominal Thk.	60.000 mm	
Surface Area		270491.2 cm²	SG of Contents	0.000	
Element Volume		17818726.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		12315.5 kgf	Operating Weight	12315.5 kgf	
			Comp. All. Ope. Stress	827.0 kgf/cm²	

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (5932 \cdot 1376) / (1499 \cdot 1 - 0.6 \cdot 5932)$
 $= 56.9044 + 1.0000 = 57.9044 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 Less Operating Hydrostatic Head Pressure of 2.340 kPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1499 \cdot 1 \cdot 59) / (1376 + 0.6 \cdot 59)$
 $= 6145 - 2.34 = 6142 \text{ kPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1509 \cdot 1 \cdot 60) / (1375 + 0.6 \cdot 60)$
 $= 6293 \text{ kPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$
 $= (5932 \cdot (1376 + 0.6 \cdot 59)) / (1 \cdot 59)$
 $= 1447.026 \text{ kgf/cm}^2$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: S-3



Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*524)/(3*48.6) = 1408 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*106)/(3*134) = 103 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463)/(3*48.6) = 1245 \text{ kPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: S-4

Cylindrical Shell Design Information:					
Design Pressure	P	5931.830 kPa	Int. Design Temperature	430 °C	
Material		SA-387 11	Ext. Design Temperature	200 °C	
Length		3000.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope) S	1499.0	kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1509.1	kgf/cm ²
Inside Diameter	Di	2750.000 mm	Actual Stress	1447.0	kgf/cm ²
Spec. Min. Thk t _e , or sa		60.0000 mm	Spec. Nominal Thk.	60.000	mm
Surface Area		270491.2 cm ²	SG of Contents	0.000	
Element Volume		17818726.0 cm ³	Weight of Contents	0.0	kgf
Empty Weight		12315.5 kgf	Operating Weight	12315.5	kgf
			Comp. All. Ope. Stress	827.0	kgf/cm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5932 \cdot 1376) / (1499 \cdot 1 - 0.6 \cdot 5932)$$

$$= 56.9044 + 1.0000 = 57.9044 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 2.340 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1499 \cdot 1 \cdot 59) / (1376 + 0.6 \cdot 59)$$

$$= 6145 - 2.34 = 6142 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1509 \cdot 1 \cdot 60) / (1375 + 0.6 \cdot 60)$$

$$= 6293 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (5932 \cdot (1376 + 0.6 \cdot 59)) / (1 \cdot 59)$$

$$= 1447.026 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: S-4



Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*524)/(3*48.6) = 1408 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*106)/(3*134) = 103 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463)/(3*48.6) = 1245 \text{ kPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: S-5

Cylindrical Shell Design Information:				
Design Pressure	P	5931.830 kPa	Int. Design Temperature	430 °C
Material		SA-387 11	Ext. Design Temperature	200 °C
Length		3000.000 mm	External Pressure Chart	CS-3
			UNS Number	K11789
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope) S	1499.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1509.1 kgf/cm²
Inside Diameter	Di	2750.000 mm	Actual Stress	1447.0 kgf/cm²
Spec. Min. Thk t _e , or sa		60.0000 mm	Spec. Nominal Thk.	60.000 mm
Surface Area		270491.2 cm²	SG of Contents	0.000
Element Volume		17818726.0 cm³	Weight of Contents	0.0 kgf
Empty Weight		12315.5 kgf	Operating Weight	12315.5 kgf
			Comp. All. Ope. Stress	827.0 kgf/cm²

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5932 \cdot 1376) / (1499 \cdot 1 - 0.6 \cdot 5932)$$

$$= 56.9044 + 1.0000 = 57.9044 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
[Less Operating Hydrostatic Head Pressure of 2.340 kPa](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1499 \cdot 1 \cdot 59) / (1376 + 0.6 \cdot 59)$$

$$= 6145 - 2.34 = 6142 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1509 \cdot 1 \cdot 60) / (1375 + 0.6 \cdot 60)$$

$$= 6293 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (5932 \cdot (1376 + 0.6 \cdot 59)) / (1 \cdot 59)$$

$$= 1447.026 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: S-5



Cylindrical Shell From 90 to 100 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*524)/(3*48.6) = 1408 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*106)/(3*134) = 103 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463)/(3*48.6) = 1245 \text{ kPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: S-6

Cylindrical Shell Design Information:					
Design Pressure	P	5931.830 kPa	Int. Design Temperature	430 °C	
Material		SA-387 11	Ext. Design Temperature	200 °C	
Length		3000.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope) S	1499.0 kgf/cm²	
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1509.1 kgf/cm²	
Inside Diameter	Di	2750.000 mm	Actual Stress	1447.0 kgf/cm²	
Spec. Min. Thk t _e , or s _a		60.0000 mm	Spec. Nominal Thk.	60.000 mm	
Surface Area		270491.2 cm²	SG of Contents	0.000	
Element Volume		17818726.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		12315.5 kgf	Operating Weight	12315.5 kgf	
			Comp. All. Ope. Stress	827.0 kgf/cm²	

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5932 \cdot 1376) / (1499 \cdot 1 - 0.6 \cdot 5932)$$

$$= 56.9044 + 1.0000 = 57.9044 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

[Less Operating Hydrostatic Head Pressure of 2.340 kPa](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1499 \cdot 1 \cdot 59) / (1376 + 0.6 \cdot 59)$$

$$= 6145 - 2.34 = 6142 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1509 \cdot 1 \cdot 60) / (1375 + 0.6 \cdot 60)$$

$$= 6293 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (5932 \cdot (1376 + 0.6 \cdot 59)) / (1 \cdot 59)$$

$$= 1447.026 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: S-6



Cylindrical Shell From 100 to 110 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*524)/(3*48.6) = 1408 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*106)/(3*134) = 103 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463)/(3*48.6) = 1245 \text{ kPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: S-7

Cylindrical Shell Design Information:				
Design Pressure	P	5931.830 kPa	Int. Design Temperature	430 °C
Material		SA-387 11	Ext. Design Temperature	200 °C
Length		1720.000 mm	External Pressure Chart	CS-3
			UNS Number	K11789
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope) S	1499.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1509.1 kgf/cm²
Inside Diameter	Di	2750.000 mm	Actual Stress	1447.0 kgf/cm²
Spec. Min. Thk t _e , or sa		60.0000 mm	Spec. Nominal Thk.	60.000 mm
Surface Area		155081.6 cm²	SG of Contents	0.000
Element Volume		10216071.0 cm³	Weight of Contents	0.0 kgf
Empty Weight		7060.9 kgf	Operating Weight	7060.9 kgf
			Comp. All. Ope. Stress	827.0 kgf/cm²

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5932 \cdot 1376) / (1499 \cdot 1 - 0.6 \cdot 5932)$$

$$= 56.9044 + 1.0000 = 57.9044 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 2.340 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1499 \cdot 1 \cdot 59) / (1376 + 0.6 \cdot 59)$$

$$= 6145 - 2.34 = 6142 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1509 \cdot 1 \cdot 60) / (1375 + 0.6 \cdot 60)$$

$$= 6293 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (5932 \cdot (1376 + 0.6 \cdot 59)) / (1 \cdot 59)$$

$$= 1447.026 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: S-7



Cylindrical Shell From 110 to 120 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*524)/(3*48.6) = 1408 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*106)/(3*134) = 103 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463)/(3*48.6) = 1245 \text{ kPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: H-2

Elliptical Head Design Information:					
Design Pressure	P	5931.830 kPa	Int. Design Temperature	430 °C	
Material		SA-387 11	Ext. Design Temperature	200 °C	
Length		50.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1499.0 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm ²
Inside Diameter	Di	2750.000 mm	Actual Stress	1464.9 kgf/cm ²	
Spec. Min. Thk t _e , or s _a		58.0000 mm	Spec. Nominal Thk.	68.000 mm	
Surface Area		94818.4 cm ²	SG of Contents	0.000	
Element Volume		3019284.5 cm ³	Weight of Contents	0.0 kgf	
Empty Weight		5242.3 kgf	Operating Weight	5242.3 kgf	
			Comp. All. Ope. Stress	822.7 kgf/cm ²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 34.375 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.188 mm

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (5932 \cdot 2752 \cdot 1) / (2 \cdot 1499 \cdot 1 - 0.2 \cdot 5932)$$

$$= 55.6976 + 1.0000 = 56.6976 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 2.340 kPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1499 \cdot 1 \cdot 57) / (1 \cdot 2752 + 0.2 \cdot 57)$$

$$= 6070 - 2.34 = 6068 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1509 \cdot 1 \cdot 58) / (1 \cdot 2750 + 0.2 \cdot 58)$$

$$= 6216 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (5932 \cdot (1 \cdot 2752 + 0.2 \cdot 57)) / (2 \cdot 1 \cdot 57)$$

$$= 1464.868 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (5932 \cdot 1376) / (1499 \cdot 1 - 0.6 \cdot 5932) + 1$$

$$= 57.904 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 2.340 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1499 \cdot 1 \cdot 67) / (1376 + 0.6 \cdot 67)$$

$$= 6954 - 2.34 = 6952 \text{ kPa}$$



Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (2752 / (2 \cdot 689))^2) / 6$
 $= 0.999032$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 10.169 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 34.375 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.188 mm

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: H-2

Elliptical Head From 120 to 130 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
57.000	2866.00	50.28	0.0027623	1000.96

$$\text{MAEP} = B / (K0 \cdot Do / t) = 1001 / (0.9 \cdot 50.3) = 2169 \text{ kPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
7.541	2866.00	380.05	0.0003655	360.75

$$\text{MAEP} = B / (K0 \cdot Do / t) = 361 / (0.9 \cdot 380) = 103 \text{ kPa}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7941.165 kPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7760.377 kPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	8054.110 kPa
UG100	= 1.10 * M.A.W.P. * Sa/S	6719.447 kPa
PED	= max(1.43*DP, 1.25*DP*ratio)	8479.170 kPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8479.170 kPa
App 27-4	= M.A.W.P.	6067.625 kPa

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm² & kPa):

From To	Stress	Allowable	Ratio	Pressure
H-1	1945.1	2844.9	0.684	8081.07
S-1	1937.9	2844.9	0.681	8081.07
S-2	1937.9	2844.9	0.681	8081.07
S-3	1937.9	2844.9	0.681	8081.07
S-4	1937.9	2844.9	0.681	8081.07
S-5	1937.9	2844.9	0.681	8081.07
S-6	1937.9	2844.9	0.681	8081.07
S-7	1937.9	2844.9	0.681	8081.07
H-2	1961.8	2844.9	0.690	8081.07

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D-10"	Nozzle	1509.10	1498.98	1.007
MH-3-24"	Nozzle	1509.10	1498.98	1.007
MH-3-24"	Pad	1509.10	1498.98	1.007
E-8"	Nozzle	1203.20	1009.30	1.192
E-8"	Pad	1509.10	1498.98	1.007
UC-2-14"	Nozzle	1509.10	1498.98	1.007
UC-2-14"	Pad	1509.10	1498.98	1.007
B-14"	Nozzle	1509.10	1498.98	1.007
B-14"	Pad	1509.10	1498.98	1.007
MH-2-24"	Nozzle	1509.10	1498.98	1.007
MH-2-24"	Pad	1509.10	1498.98	1.007
UC-1-14"	Nozzle	1509.10	1498.98	1.007
UC-1-14"	Pad	1509.10	1498.98	1.007
A-10"	Nozzle	1203.20	1009.30	1.192
A-10"	Pad	1509.10	1498.98	1.007
MH-1-24"	Nozzle	1509.10	1498.98	1.007
MH-1-24"	Pad	1509.10	1498.98	1.007
C-14"	Nozzle	1509.10	1498.98	1.007
C-14"	Pad	1509.10	1498.98	1.007

Minimum

1.007

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
H-1	1509.10	1498.98	1.007
S-1	1509.10	1498.98	1.007
S-2	1509.10	1498.98	1.007
S-3	1509.10	1498.98	1.007
S-4	1509.10	1498.98	1.007
S-5	1509.10	1498.98	1.007
S-6	1509.10	1498.98	1.007



S-7	1509.10	1498.98	1.007
H-2	1509.10	1498.98	1.007

Minimum 1.007

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
D-10"	819.24	2844.86	0.288
MH-3-24"	804.28	2844.86	0.283
E-8"	415.32	1881.28	0.221
UC-2-14"	781.02	2844.86	0.275
B-14"	781.02	2844.86	0.275
MH-2-24"	804.28	2844.86	0.283
UC-1-14"	781.02	2844.86	0.275
A-10"	417.00	1881.28	0.222
MH-1-24"	804.28	2844.86	0.283
C-14"	882.77	2844.86	0.310

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b) = 1.30 * M.A.W.P. * Sa/S 7941.165 kPa
 UG99b(b)(35) = 1.30 * Design Pres * Sa/S 7760.377 kPa
 UG99(c) = 1.30 * M.A.P. - Head(Hyd) 8073.836 kPa
 UG100 = 1.10 * M.A.W.P. * Sa/S 6719.447 kPa
 PED = max(1.43*DP, 1.25*DP*ratio) 8479.170 kPa
 PED&UG99(b)(35) = max(1.43*DP, 1.3*DP*ratio) 8479.170 kPa
 App 27-4 = M.A.W.P. 6067.625 kPa

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 5929 • 1.01
 = 7760.377 kPa

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm² & kPa):

From To	Stress	Allowable	Ratio	Pressure
H-1	1950.7	2844.9	0.686	7968.13
S-1	1942.0	2844.9	0.683	7960.90
S-2	1934.8	2844.9	0.680	7931.50
S-3	1927.7	2844.9	0.678	7902.09
S-4	1920.5	2844.9	0.675	7872.68
S-5	1913.3	2844.9	0.673	7843.28
S-6	1906.1	2844.9	0.670	7813.87
S-7	1899.0	2844.9	0.668	7784.47
H-2	1918.2	2844.9	0.674	7767.61

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D-10"	Nozzle	1509.10	1498.98	1.007
MH-3-24"	Nozzle	1509.10	1498.98	1.007
MH-3-24"	Pad	1509.10	1498.98	1.007
E-8"	Nozzle	1203.20	1009.30	1.192
E-8"	Pad	1509.10	1498.98	1.007
UC-2-14"	Nozzle	1509.10	1498.98	1.007
UC-2-14"	Pad	1509.10	1498.98	1.007
B-14"	Nozzle	1509.10	1498.98	1.007
B-14"	Pad	1509.10	1498.98	1.007
MH-2-24"	Nozzle	1509.10	1498.98	1.007
MH-2-24"	Pad	1509.10	1498.98	1.007
UC-1-14"	Nozzle	1509.10	1498.98	1.007
UC-1-14"	Pad	1509.10	1498.98	1.007
A-10"	Nozzle	1203.20	1009.30	1.192
A-10"	Pad	1509.10	1498.98	1.007
MH-1-24"	Nozzle	1509.10	1498.98	1.007
MH-1-24"	Pad	1509.10	1498.98	1.007
C-14"	Nozzle	1509.10	1498.98	1.007
C-14"	Pad	1509.10	1498.98	1.007

Minimum 1.007

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
H-1	1509.10	1498.98	1.007
S-1	1509.10	1498.98	1.007



S-2	1509.10	1498.98	1.007
S-3	1509.10	1498.98	1.007
S-4	1509.10	1498.98	1.007
S-5	1509.10	1498.98	1.007
S-6	1509.10	1498.98	1.007
S-7	1509.10	1498.98	1.007
H-2	1509.10	1498.98	1.007

Minimum

1.007

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
D-10"	807.79	2844.86	0.284
MH-3-24"	792.32	2844.86	0.279
E-8"	409.14	1881.28	0.217
UC-2-14"	766.57	2844.86	0.269
B-14"	766.57	2844.86	0.269
MH-2-24"	783.54	2844.86	0.275
UC-1-14"	758.04	2844.86	0.266
A-10"	403.21	1881.28	0.214
MH-1-24"	774.76	2844.86	0.272
C-14"	848.53	2844.86	0.298

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	6100	...	...	No Calc	No Calc
20	30	6100	...	...	No Calc	No Calc
30	40	6100	...	...	No Calc	No Calc
40	50	No Calc	2867	57.5	0.0027855	1002.4
50	60	20278.3	2870	59	0.00053048	523.646
60	70	20278.3	2870	59	0.00053048	523.646
70	80	20278.3	2870	59	0.00053048	523.646
80	90	20278.3	2870	59	0.00053048	523.646
90	100	20278.3	2870	59	0.00053048	523.646
100	110	20278.3	2870	59	0.00053048	523.646
110	120	20278.3	2870	59	0.00053048	523.646
120	130	No Calc	2866	57	0.0027623	1000.96

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kPa	External M.A.W.P. kPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	58.5	8.5438	103.421	2191
50	60	60	22.4342	103.421	1408
60	70	60	22.4342	103.421	1408
70	80	60	22.4342	103.421	1408
80	90	60	22.4342	103.421	1408
90	100	60	22.4342	103.421	1408
100	110	60	22.4342	103.421	1408
110	120	60	22.4342	103.421	1408
120	130	58	8.54117	103.421	2169

Minimum

1408

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	6100	No Calc	No Calc	No Calc
20	30	6100	No Calc	No Calc	No Calc
30	40	6100	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	20278.3	902456	No Calc	No Calc
60	70	20278.3	902456	No Calc	No Calc
70	80	20278.3	902456	No Calc	No Calc
80	90	20278.3	902456	No Calc	No Calc
90	100	20278.3	902456	No Calc	No Calc
100	110	20278.3	902456	No Calc	No Calc
110	120	20278.3	902456	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm³	Corroded Metal Wgt. kg	Corroded ID Volume cm³	Extra due Misc % kg
10	20	5089.83	...	4834.05	...	...
20	30	3455.34	...	3148.18	...	...
30	40	691.068	...	629.635	...	...
40	50	5242.32	3019284	5165.23	3025660	...
50	60	12315.5	17818726	12114.5	17844652	...
60	70	12315.5	17818726	12114.5	17844652	...
70	80	12315.5	17818726	12114.5	17844652	...
80	90	12315.5	17818726	12114.5	17844652	...
90	100	12315.5	17818726	12114.5	17844652	...
100	110	12315.5	17818726	12114.5	17844652	...
110	120	7060.87	10216071	6945.67	10230935	...
120	130	5242.32	3019284	5165.23	3025660	...
Total		100674	0.123E+09	98575	0.123E+09	0

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Plat	982.951	1471.96	2100	-1471.96	PL-6
10	Insl	846.982	...	1250	...	FR-3
10	Lini	802.999	...	1250	...	FR-3.1
10	Wght	1000	...	...	...	Taling Lug
20	Insl	1016.38	...	1500	...	FR-2
20	Lini	965.107	...	1500	...	FR-2.1
20	Wght	500	...	0.00028956	...	Skirt MW
30	Insl	203.276	...	300	...	FR-1
30	Lini	193.021	...	300	...	FR-1.1
40	Liqd	2309.12	...	-343.75	...	Liquid 10
40	Insl	655.82	...	-318.75	...	Ins: 10
40	Lini	227.88	...	-318.75	...	clad
40	Nozl	466.917	...	-833.256	...	D-10"
40	Wght	196	2840	-685	...	PP_D
50	Plat	3931.29	0.21088E-04	0.48768E-04	...	PL-1
50	Liqd	13654.5	...	1500	...	Liquid 20
50	Insl	1502.85	...	1500	...	Ins: 20
50	Lini	623.457	...	1500	...	clad
50	Nozl	1453.43	-1625	1500	...	MH-3-24"
50	Nozl	396.697	1625	500	...	E-8"
50	Wght	1349	2815	500	...	PP_E
60	Plat	2039.64	...	1610	1516.12	PL-2
60	Pack	1063.14	...	2412.5	...	Bed#2
60	Liqd	682.724	...	75	...	Liquid: 30
60	Insl	1502.85	...	1500	...	Ins: 30
60	Lini	623.457	...	1500	...	clad
60	Nozl	479.063	1625	2200	...	UC-2-14"
60	Nozl	324.334	1625	600	...	B-14"
60	Wght	630	...	1825	...	DL_BS-1
60	Wght	2788	2891	600	...	PP_B
70	Plat	2039.64	...	2610	-1516.12	PL-3
70	Pack	2714.41	...	1500	...	Bed#2
70	Insl	1502.85	...	1500	...	Ins: 40
70	Lini	623.457	...	1500	...	clad
80	Pack	294.061	...	162.5	...	Bed#2
80	Pack	764.558	...	2577.5	...	Bed#1
80	Insl	1502.85	...	1500	...	Ins: 50
80	Lini	623.457	...	1500	...	clad
80	Nozl	1453.43	1625	1500	...	MH-2-24"
80	Wght	630	...	2155	...	DL_BS-2
80	Wght	800	...	799.999	...	DL_LD-2



80 Wght	160	...	325	...	DL_BL-2
80 Wght	1100	...	799.999	...	OP_LD-2
80 Wght	100	...	770	...	TA_LD-2
80 Wght	160	...	950	...	small Noz
80 Wght	4252	...	950	...	Cleat Weight
80 Wght	10000	...	950	...	Margin Weight
90 Plat	2039.64	...	610.001	1516.12	PL-4
90 Pack	2714.41	...	1500	...	Bed#1
90 Insl	1502.85	...	1500	...	Ins: 60
90 Lini	623.457	...	1500	...	clad
90 Nozl	479.063	1625	500	...	UC-1-14"
100 Plat	2039.64	...	1610	-1516.12	PL-5
100 Pack	592.646	...	327.5	...	Bed#1
100 Insl	1502.85	...	1500	...	Ins: 70
100 Lini	623.457	...	1500	...	clad
100 Nozl	311.064	1625	1850	...	A-10"
100 Wght	800	...	1110	...	DL_LD-1
100 Wght	160	...	655.001	...	DL_BL-1
100 Wght	250	...	1850	...	DL_FP
100 Wght	1100	...	1110	...	OP_LD-1
100 Wght	140	...	1850	...	OP_FP
100 Wght	100	...	1200	...	TA_LD-1
100 Wght	3535	2840	1850	...	PP_A
110 Insl	861.635	...	860	...	Ins: 80
110 Lini	357.449	...	860	...	clad
110 Nozl	1453.43	1625	860	...	MH-1-24"
110 Wght	230	...	1350	...	DL_PAD
110 Wght	200	...	1350	...	TA_PAD
120 Plat	8482.56	...	890.001	...	TOP PL
120 Insl	661.372	...	368.75	...	Ins: 90
120 Lini	227.88	...	368.75	...	clad
120 Nozl	377.955	...	958.5	...	C-14"
120 Wght	3578	2891	687	...	PP_C
120 Wght	2000	3500	642	...	PSV
120 Wght	1500	...	...	...	Lifting Lug
120 Wght	600	1400	...	...	Pipe Davit
120 Forc	...	...	685	...	C_FxZ
60 Pliq	1342.85	...	2412.500	...	
70 Pliq	3428.57	...	1500.000	...	
80 Pliq	371.43	...	162.500	...	
80 Pliq	965.71	...	2577.500	...	
90 Pliq	3428.57	...	1500.000	...	
100 Pliq	748.57	...	327.500	...	

Total Weight of Each Detail Type:

Platforms	21555.4
Packing	8143.2
Packing Liquid	10285.7
Liquid	16646.3
Insulation	13262.6
Lining	6515.1
Nozzles	7195.4
Weights	37858.0

Sum of the Detail Weights 121461.6 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	100674.6	100674.6	100674.6	100674.6	100674.6	100674.6
Nozzles	7195.4	7195.4	7195.4	7195.4	7195.4	7195.4
Wld Weights	18412.0	18412.0	18412.0	18412.0	18412.0	18412.0
Platforms	...	...	21555.4	21555.4	21555.4	21555.4
Packing	...	...	...	...	8143.2	8143.2
Insulation	...	...	13262.6	13262.6	13262.6	13262.6
Lining	6515.1	6515.1	6515.1	6515.1	6515.1	6515.1

*



Empty Weights	...	...	...	5660.0	5660.0	...
Test Weights	...	5660.0	...	...	...	...
Ope Weights	...	...	...	...	...	19446.0
Ope. Liquid	...	...	...	...	...	16646.3
Packing Liq	...	...	...	...	...	10285.7
Test Liquid	...	122544.6	...	...	...	...
Totals	132797.1	261001.7	167615.0	173275.0	181418.2	222136.3

Field Installation Options:

* Packing installed after lifting.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	132797.1 kg
Shop Test Wt.	- Fabricated weight + water (full)	261001.7 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	167615.0 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	173275.0 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	181418.2 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	222136.3 kg
Field Test Wt.	- Empty weight + test medium (full)	295819.6 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	86069.4 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	123676.4 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	73226.0 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	207816	204675
20	30	249380	245987
30	40	49875.9	49197.4
40	50	94818.4	86296.1
50	60	270491	259181
60	70	270491	259181
70	80	270491	259181
80	90	270491	259181
90	100	270491	259181
100	110	270491	259181
110	120	155082	148597
120	130	94818.4	86296.1

Total	2474737.5	2376136.8 cm ²
Total	2663.8	2557.7 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	8722.76	8722.76	8722.76	1446.87	8722.76
20	30	5936.83	5936.83	5936.83	...	5936.83
30	40	1087.37	1087.37	1087.37	...	1087.37
40	50	6592.94	9098.06	9610.87	556.641	6788.94
50	60	20223.2	35226.7	38033.9	2080.25	21572.2
60	70	17914.8	23791.5	35725.6	9365.64	21766
70	80	16481.4	22624.4	34292.2	...	19195.8
80	90	31997.2	35493	49807.9	2361.83	34155.8
90	100	16960.5	23103.5	34771.2	778.478	19674.9
100	110	18102.5	24218.7	35913.2	10544.9	23470.1
110	120	10163.4	10163.4	20374.9	2361.83	10163.4
120	130	19092.1	22670.1	22110	18184	22670.1



Empty Support Force + the Sum of the Y forces 166521.42 kgf
Operating Support Force + the Sum of the Y forces 225668.38 kgf
Hydro Support Force + the Sum of the Y forces 299918.84 kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg	Cumulative Oper. Wgt. kg	Cumulative Hydro. Wgt. kg
10	20	195204	222136	296387
20	30	186481	213414	287664
30	40	180545	207477	281727
40	50	179457	206389	280640
50	60	172668	197291	271029
60	70	151096	162065	232995
70	80	129330	138273	197269
80	90	110134	115649	162977
90	100	75978.5	80155.6	113169
100	110	56303.6	57052.2	78398.1
110	120	32833.5	32833.5	42484.9
120	130	22670.1	22670.1	22110

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	47680.4	47680.4	47680.4
20	30	46233.6	46233.6	46233.6
30	40	46233.6	46233.6	46233.6
40	50	46233.6	46233.6	46233.6
50	60	45676.9	45676.9	45676.9
60	70	43596.7	43596.7	43596.7
70	80	34231	34231	34231
80	90	34231	34231	34231
90	100	31869.2	31869.2	31869.2
100	110	31090.7	31090.7	31090.7
110	120	20545.8	20545.8	20545.8
120	130	18184	18184	18184



Nozzle Flange MAWP Results: (kPa & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
D-10"	7278.1	10340.0	430	600	GR 1.9	...	...	...	...
MH-3-24"	7278.1	10340.0	430	600	GR 1.9	...	...	...	...
E-8"	7278.1	10340.0	430	600	GR 1.9	...	...	...	...
UC-2-14"	7278.1	10340.0	430	600	GR 1.9	...	...	...	...
B-14"	7278.1	10340.0	430	600	GR 1.9	...	...	...	...
MH-2-24"	7278.1	10340.0	430	600	GR 1.9	...	...	...	...
UC-1-14"	7278.1	10340.0	430	600	GR 1.9	...	...	...	...
A-10"	7278.1	10340.0	430	600	GR 1.9	...	...	...	...
MH-1-24"	7278.1	10340.0	430	600	GR 1.9	...	...	...	...
C-14"	10914.1	15510.0	430	900	GR 1.9	...	...	...	...
Min Rating	7278.1	10340.0, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 1.43489 Hz.

The Natural Frequency for the Vessel (Ope...) is 1.39797 Hz.

The Natural Frequency for the Vessel (Filled) is 1.20519 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	3607.52	6355.57

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	3607.52	6355.57

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
120	685.00	2551.	-3532.	2551.	2747.		2747.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
= $\min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$
= $\min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.91)/(1 + 1.7 \cdot 3.4 \cdot 0.19)))$
= $\min(0.85, 0.88)$
= 0.850

Natural Frequency of Vessel (Operating)	1.398 Hz
Natural Frequency of Vessel (Empty)	1.435 Hz
Natural Frequency of Vessel (Test)	1.205 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.610
Structure Height to Diameter ratio	8.790
Height to top of Structure	26815.500 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Z_g = 274320$ mm

Effective Height $[z]$:
= Centroid Height + Vessel Base Elevation
= $1250 + 300 = 1550$ mm
= 5.09 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
Because z (5.09 ft.) < 15 ft.
= $2.01(15 / Z_g)^{2/\alpha}$
= $2.01(15/900)^{2/9.5}$
= 0.85

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
= 1, per Table 26.6-1



As there is No Hill Present: [Kzt]:
 $K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:
 $= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0 \cdot 0 \cdot 0)^2$
 $= 1$

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 $= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$
 $= 16 \text{ psf } [78.1 \text{ kgf/m}^2]$

Force on the first element [F]:
 $= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 16 \cdot 0.85 \cdot 0.61 \cdot 88.7$
 $= 736 \text{ lbs. } [334 \text{ kgf}]$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
K-3	1550.0	0.000	0.000	0.000	0.849	1.000	78.119
K-2	4300.0	0.000	0.000	0.000	0.849	1.000	78.119
K-1	6100.0	0.000	0.000	0.000	0.902	1.000	78.119
H-1	6942.6	0.000	0.000	0.000	0.927	1.000	78.119
S-1	7950.0	0.000	0.000	0.000	0.954	1.000	78.119
S-2	10950.0	0.000	0.000	0.000	1.020	1.000	78.119
S-3	13950.0	0.000	0.000	0.000	1.074	1.000	78.119
S-4	16950.0	0.000	0.000	0.000	1.119	1.000	78.119
S-5	19950.0	0.000	0.000	0.000	1.158	1.000	78.119
S-6	22950.0	0.000	0.000	0.000	1.192	1.000	78.119
S-7	25310.0	0.000	0.000	0.000	1.217	1.000	78.119
H-2	26573.2	0.000	0.000	0.000	1.230	1.000	78.119

Note:
The height/(avg. diameter) ratio [8.4], was less than or equal to 15,
therefore no wind vibration calculations have been performed.

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm	Pressure kgf/m ²	Force kgf	Cf
PL-6	23128.00	2400.00	78.12	180.67	1.00
PL-1	30000.00	6450.00	78.12	234.36	1.00
PL-2	30000.00	11060.00	78.12	234.36	1.00
PL-3	30000.00	15060.00	78.12	234.36	1.00
PL-4	30000.00	19060.00	78.12	234.36	1.00
PL-5	30000.00	23060.00	78.12	234.36	1.00
TOP PL	70000.00	27060.00	78.12	546.83	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure kgf/m ²	Force kgf
Taling Lug	0.00	300.00	78.12	0.00
Skirt MW	0.00	2800.00	78.12	0.00
PPD	0.00	5715.00	78.12	0.00
PPE	0.00	6950.00	78.12	0.00
DLBS-1	0.00	11275.00	78.12	0.00
PPB	0.00	10050.00	78.12	0.00
DLBS-2	0.00	17605.00	78.12	0.00
DLLD-2	0.00	16250.00	78.12	0.00
DLBL-2	0.00	15775.00	78.12	0.00
OPLD-2	0.00	16250.00	78.12	0.00
TALD-2	0.00	16220.00	78.12	0.00



small Noz	0.00	16400.00	78.12	0.00
Cleat Weight	0.00	16400.00	78.12	0.00
Margin Weight	0.00	16400.00	78.12	0.00
DLID-1	0.00	22560.00	78.12	0.00
DLBL-1	0.00	22105.00	78.12	0.00
DLFP	0.00	23300.00	78.12	0.00
OPLD-1	0.00	22560.00	78.12	0.00
OPFP	0.00	23300.00	78.12	0.00
TALD-1	0.00	22650.00	78.12	0.00
PPA	0.00	23300.00	78.12	0.00
DLPAD	0.00	25800.00	78.12	0.00
TAPAD	0.00	25800.00	78.12	0.00
PPC	0.00	26857.00	78.12	0.00
PSV	0.00	26812.00	78.12	0.00
Lifting Lug	0.00	26170.00	78.12	0.00
Pipe Davit	0.00	26170.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 1.39797 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	1550	3295.2	82380	78.1189	674.738
20	30	4300	3295.2	98856	78.1189	440.413
30	40	6100	3295.2	19771.2	78.1189	88.0826
40	50	6942.61	3800.4	28644.3	78.1189	127.613
50	60	7950	3804	114120	78.1189	717.568
60	70	10950	3804	114120	78.1189	768.797
70	80	13950	3804	114120	78.1189	786.72
80	90	16950	3804	114120	78.1189	508.416
90	100	19950	3804	114120	78.1189	754.707
100	110	22950	3804	114120	78.1189	767.443
110	120	25310	3804	65428.8	78.1189	291.492
120	130	26573.2	3799.2	28620.3	78.1189	740.041

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(87.30.75)$
 $= 0.571$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/1.4)$
 $= 0.715$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.57, 0.72)$ per 12.8.2
 $= 0.715$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R / I)$
 $= 0.16 / (1/1)$
 $= 0.157$

Maximum value of Cs per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R / Ie))$
 $= 0.1 / (0.72 (1/1))$
 $= 0.145$

Minimum value of Cs per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot SDS \cdot Ie, 0.030)$
 $= \max(0.044 \cdot 0.16 \cdot 1, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= Cs \cdot W$ (Equation 12.8-1):
 $= 0.15 \cdot 222136$
 $= 32296.105$ kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$



$$= 0.2 \cdot 0.16 \cdot 222136$$

$$= 6966.2 \text{ kgf}$$

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_x^k / (\sum W_i \cdot h_i^k)$ and k is an exponent which is related
to the period of Vibration.

In this case, the value of k was 1.11

The Natural Frequency for the Vessel (Ope...) is 1.39797 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight kgf	Element Ope Load kgf	Element Emp Load kgf	Element Vertical Load kgf
10	20	1250	8722.76	78.9018	81.1353	3434.09
20	30	4000	5936.83	194.77	198.151	3160.55
30	40	5800	1087.37	53.8375	54.585	2974.37
40	50	6125	9098.06	478.504	351.387	2940.27
50	60	7650	35226.7	2370.06	1376	2654.95
60	70	10650	23791.5	2309.23	1753.14	1550.24
70	80	13650	22624.4	2890.74	2118.33	804.142
80	90	16650	35493	5651.27	5115.5	94.6404
90	100	19650	23103.5	4419.52	3252.72	-1018.42
100	110	22650	24218.7	5422.48	4058.15	-1742.94
110	120	25010	10163.4	2539.6	2540.45	-2502.44
120	130	25895	22670.1	5887.19	4958.1	-2821.17



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas kgf	Cumulative Shr Eqk Cas kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	1250	3607.52	3607.52	99682.2	99682.2
20	30	4000	3607.52	3607.52	90663.4	90663.4
30	40	5800	3607.52	3607.52	79840.8	79840.8
40	50	6125	3607.52	3607.52	77676.3	77676.3
50	60	7650	3607.52	3607.52	77495.9	77495.9
60	70	10650	3607.52	3607.52	66673.3	66673.3
70	80	13650	3607.52	3607.52	55850.8	55850.8
80	90	16650	3607.52	3607.52	45028.2	45028.2
90	100	19650	3607.52	3607.52	34205.6	34205.6
100	110	22650	3607.52	3607.52	23383.1	23383.1
110	120	25010	3607.52	3607.52	12560.5	12560.5
120	130	25895	3607.52	3607.52	6355.57	6355.57

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	1250	6666.03	32296.1	95724.6	608251
20	30	4000	5991.29	32217.2	79903	527610
30	40	5800	5550.88	32022.4	62589.7	431250
40	50	6125	5462.8	31968.6	59285.6	412053
50	60	7650	5335.18	31490.1	59015.6	410466
60	70	10650	4617.62	29120	44086.4	319551
70	80	13650	3848.82	26810.8	31386.8	235655
80	90	16650	3062.1	23920.1	21020.4	159558
90	100	19650	2553.68	18268.8	12596.7	96274.9
100	110	22650	1798.98	13849.3	6067.69	48097.7
110	120	25010	1031.53	8426.79	1821.92	14683.6
120	130	25895	740.041	5887.19	298.367	2373.57

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	3607.52	0.027305	0.10755	0.84701E-04
20	30	3607.52	0.27424	0.51697	0.0001861
30	40	3607.52	0.57421	0.63424	0.00020471
40	50	3607.52	0.63936	0.64449	0.00020509
50	60	3607.52	0.9601	1.29093	0.00022536
60	70	3607.52	1.63582	1.99363	0.0002426
70	80	3607.52	2.36323	2.74347	0.00025679
80	90	3607.52	3.13321	3.53131	0.00026793
90	100	3607.52	3.93662	4.34801	0.00027603
100	110	3607.52	4.76433	5.18445	0.00028109
110	120	3607.52	5.42658	5.66937	0.00028261
120	130	3607.52	5.67644	5.6835	0.00028261

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
Allowable deflection : 129.600 Actual deflection : 5.684 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	6666.03	0.026189	0.10174	0.79019E-04
20	30	5991.29	0.25502	0.47255	0.00016471
30	40	5550.88	0.52311	0.57595	0.00017983
40	50	5462.8	0.58045	0.58495	0.00018016
50	60	5335.18	0.86205	1.15164	0.00019682
60	70	4617.62	1.45196	1.76141	0.000209
70	80	3848.82	2.0785	2.40191	0.00021744
80	90	3062.1	2.73044	3.06302	0.00022285
90	100	2553.68	3.39867	3.73653	0.00022581
100	110	1798.98	4.07589	4.41614	0.00022703
110	120	1031.53	4.61145	4.80684	0.00022722
120	130	740.041	4.81252	4.8182	0.00022722

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed m/sec	2nd Crit. Wind Speed m/sec
10	20	22.971	143.569
20	30	22.971	143.569
30	40	22.971	143.569
40	50	26.4928	165.58
50	60	26.5179	165.737
60	70	26.5179	165.737
70	80	26.5179	165.737
80	90	26.5179	165.737
90	100	26.5179	165.737
100	110	26.5179	165.737
110	120	26.5179	165.737
120	130	26.4844	165.528



Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 129.600 Actual deflection : 4.818 mm

Total Deflection in the Operating Condition + Applied Forces:
Allowable deflection : 129.600 Actual deflection : 10.502 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	1649.97	1517.96	107507792	98965512
20	30	1486.1	1353.99	96977080	88409008
30	40	1486.1	1353.99	96977080	88409008
40	50	5161.55	5075.13	355164800	349461312
50	60	5296.73	5210.3	364482080	358784544
60	70	5296.73	5210.3	364482080	358784544
70	80	5296.73	5210.3	364482080	358784544
80	90	5296.73	5210.3	364482080	358784544
90	100	5296.73	5210.3	364482080	358784544
100	110	5296.73	5210.3	364482080	358784544
110	120	5296.73	5210.3	364482080	358784544
120	130	5116.53	5030.1	352061632	346356128



Longitudinal Allowable Stresses:

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	1181.99	2004.25	-1229.73	-1229.73
20	30	1181.99	2004.25	-1194.74	-1194.74
30	40	905.46	1706.92	-1068.8	-1375.58
40	50	1798.78	3413.83	-823.769	-1463.4
50	60	1798.78	3413.83	-827.029	-1465.25
60	70	1798.78	3413.83	-827.029	-1465.25
70	80	1798.78	3413.83	-827.029	-1465.25
80	90	1798.78	3413.83	-827.029	-1465.25
90	100	1798.78	3413.83	-827.029	-1465.25
100	110	1798.78	3413.83	-827.029	-1465.25
110	120	1798.78	3413.83	-827.029	-1465.25
120	130	1798.78	3413.83	-822.662	-1462.78

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	711.669	-13.4152	931.049
50	60	693.268	-13.0945	906.975
60	70	693.268	-13.0945	906.975
70	80	693.268	-13.0945	906.975
80	90	693.268	-13.0945	906.975
90	100	693.268	-13.0945	906.975
100	110	693.268	-13.0945	906.975
110	120	693.268	-13.0945	906.975
120	130	718.018	-13.5258	939.355

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	-128.597	-146.339	-195.254	48.1789	48.1789
20	30	-137.727	-157.618	-212.456	52.2951	52.2951
30	40	-133.343	-153.233	-208.072	52.2951	52.2951
40	50	-35.3602	-37.3869	-35.3602	13.23	13.23
50	60	-33.1399	-35.114	-33.1399	12.731	12.731
60	70	-28.9996	-30.9737	-28.9996	12.1512	12.1512
70	80	-24.8221	-26.5384	-24.8221	9.54084	9.54084
80	90	-21.1378	-22.1962	-21.1378	9.54084	9.54084
90	100	-14.5824	-15.3841	-14.5824	8.88255	8.88255
100	110	-10.8062	-10.9499	-10.8062	8.66557	8.66557
110	120	-6.30166	-6.30166	-6.30166	5.72651	5.72651
120	130	-4.50689	-4.50689	-4.50689	5.25009	5.25009

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	48.1789	96.7253	614.609	48.3626	...
20	30	52.2951	90.3788	596.783	45.1894	...
30	40	52.2951	70.7956	487.79	35.3978	...
40	50	13.23	16.9649	117.911	8.48243	...
50	60	12.731	16.4488	114.405	8.22439	...
60	70	12.1512	12.2877	89.0649	6.14386	...
70	80	9.54084	8.74808	65.6814	4.37404	...
80	90	9.54084	5.85877	44.4719	2.92939	...
90	100	8.88255	3.51093	26.8336	1.75547	...
100	110	8.66557	1.69118	13.4057	0.84559	...
110	120	5.72651	0.5078	4.0926	0.2539	...
120	130	5.25009	0.086144	0.6853	0.043072	...



Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case kgf/cm ²	Vortex Shedding Empty Case kgf/cm ²	Vortex Shedding Test Case kgf/cm ²	Earthquake Empty Case kgf/cm ²
10	20	...	...	...	501.472
20	30	...	...	...	488.345
30	40	...	...	...	401.213
40	50	...	...	...	97.1147
50	60	...	...	...	94.2378
60	70	...	...	...	73.765
70	80	...	...	...	54.6004
80	90	...	...	...	37.0544
90	100	...	...	...	22.5327
100	110	...	...	...	11.5096
110	120	...	...	...	3.543
120	130	...	...	...	0.57715

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
10	20	-2.32688	2.26231
20	30	-2.60866	2.33425
30	40	-2.60866	2.19674
40	50	-0.69596	0.57935
50	60	-0.67791	0.50956
60	70	-0.67791	0.29754
70	80	-0.67791	0.15434
80	90	-0.67791	0.018164
90	100	-0.67791	-0.19546
100	110	-0.67791	-0.33452
110	120	-0.67791	-0.48029
120	130	-0.70219	-0.56086

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	100.724	100.724	92.7209	92.7209
20	30	102.55	102.55	93.4895	93.4895
30	40	90.3085	90.3085	82.3296	82.3296
40	50	22.2274	22.2274	21.8705	21.8705
50	60	21.5996	21.5996	21.2619	21.2619
60	70	18.5831	18.5831	18.2926	18.2926
70	80	15.5667	15.5667	15.3233	15.3233
80	90	12.5502	12.5502	12.354	12.354
90	100	9.53376	9.53376	9.38473	9.38473
100	110	6.5173	6.5173	6.41543	6.41543
110	120	3.50085	3.50085	3.44613	3.44613
120	130	1.83498	1.83498	1.80524	1.80524

Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
10	20	...
20	30	...
30	40	...



40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
EP = Longitudinal Stress due to External Pressure
HP = Longitudinal Stress due to Hydrotest Pressure
NP = No Pressure
EW = Longitudinal Stress due to Weight (No Liquid)
OW = Longitudinal Stress due to Weight (Operating)
HW = Longitudinal Stress due to Weight (Hydrotest)
WI = Bending Stress due to Wind Moment (Operating)
EQ = Bending Stress due to Earthquake Moment (Operating)
EE = Bending Stress due to Earthquake Moment (Empty)
HI = Bending Stress due to Wind Moment (Hydrotest)
HE = Bending Stress due to Earthquake Moment (Hydrotest)
WE = Bending Stress due to Wind Moment (Empty) (no CA)
WF = Bending Stress due to Wind Moment (Filled) (no CA)
CW = Longitudinal Stress due to Weight (Empty) (no CA)
VO = Bending Stress due to Vortex Shedding Loads (Ope)
VE = Bending Stress due to Vortex Shedding Loads (Emp)
VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
FW = Axial Stress due to Vertical Forces for the Wind Case
FS = Axial Stress due to Vertical Forces for the Seismic Case
BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	114.70	1181.99	-376.55	1229.73	0.0970	0.3062
20	104.89	1181.99	-385.56	1194.74	0.0887	0.3227
30	77.45	905.46	-349.35	1068.80	0.0855	0.3269
40	16.37	1798.78	-88.48	823.77	0.0091	0.1074
50	16.96	1798.78	-84.60	827.03	0.0094	0.1023
60	13.34	1798.78	-72.70	827.03	0.0074	0.0879
70	8.36	1798.78	-59.36	827.03	0.0046	0.0718
80	6.13	1798.78	-49.77	827.03	0.0034	0.0602
90	6.67	1798.78	-37.19	827.03	0.0037	0.0450
100	5.39	1798.78	-28.36	827.03	0.0030	0.0343
110	2.76	1798.78	-16.71	827.03	0.0015	0.0202
120	1.96	1798.78	-12.38	822.66	0.0011	0.0150

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	524.04	1181.99	-776.71	1229.73	0.4434	0.6316
20	507.80	1181.99	-778.58	1194.74	0.4296	0.6517
30	412.67	905.46	-674.96	1068.80	0.4558	0.6315
40	97.79	1798.78	-167.35	823.77	0.0544	0.2032
50	95.94	1798.78	-161.20	827.03	0.0533	0.1949
60	75.80	1798.78	-133.20	827.03	0.0421	0.1611



70	55.04	1798.78	-104.38	827.03	0.0306	0.1262
80	38.03	1798.78	-80.27	827.03	0.0211	0.0971
90	26.17	1798.78	-55.73	827.03	0.0145	0.0674
100	15.55	1798.78	-37.83	827.03	0.0086	0.0457
110	5.99	1798.78	-19.55	827.03	0.0033	0.0236
120	2.59	1798.78	-12.73	822.66	0.0014	0.0155

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	96.96	1181.99	-394.29	1229.73	0.0820	0.3206
20	85.00	1181.99	-405.45	1194.74	0.0719	0.3394
30	57.56	905.46	-369.24	1068.80	0.0636	0.3455
40	14.34	1798.78	-90.51	823.77	0.0080	0.1099
50	14.99	1798.78	-86.57	827.03	0.0083	0.1047
60	11.37	1798.78	-74.67	827.03	0.0063	0.0903
70	6.64	1798.78	-61.07	827.03	0.0037	0.0738
80	5.08	1798.78	-50.82	827.03	0.0028	0.0615
90	5.87	1798.78	-37.99	827.03	0.0033	0.0459
100	5.25	1798.78	-28.50	827.03	0.0029	0.0345
110	2.76	1798.78	-16.71	827.03	0.0015	0.0202
120	1.96	1798.78	-12.38	822.66	0.0011	0.0150

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	619.44	1181.99	-907.59	1229.73	0.5241	0.7380
20	596.34	1181.99	-906.91	1194.74	0.5045	0.7591
30	479.36	905.46	-781.43	1068.80	0.5294	0.7311
40	116.56	1798.78	-190.18	823.77	0.0648	0.2309
50	114.13	1798.78	-183.34	827.03	0.0634	0.2217
60	89.12	1798.78	-150.48	827.03	0.0495	0.1819
70	64.40	1798.78	-117.17	827.03	0.0358	0.1417
80	44.38	1798.78	-88.74	827.03	0.0247	0.1073
90	29.67	1798.78	-60.83	827.03	0.0165	0.0736
100	17.30	1798.78	-39.87	827.03	0.0096	0.0482
110	6.54	1798.78	-20.10	827.03	0.0036	0.0243
120	2.70	1798.78	-12.84	822.66	0.0015	0.0156

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-291.80	1229.73		0.2373
20		2004.25	-309.94	1194.74		0.2594
30		1706.92	-295.76	1375.58		0.2150
40		3413.83	-57.07	1463.40		0.0390
50		3413.83	-54.10	1465.25		0.0369
60		3413.83	-47.29	1465.25		0.0323
70		3413.83	-38.74	1465.25		0.0264
80		3413.83	-33.61	1465.25		0.0229
90		3413.83	-25.22	1465.25		0.0172
100		3413.83	-20.32	1465.25		0.0139
110		3413.83	-12.28	1465.25		0.0084
120	0.79	3413.83	-9.80	1462.78	0.0002	0.0067

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-243.43	1229.73		0.1980
20		2004.25	-264.75	1194.74		0.2216
30		1706.92	-260.37	1375.58		0.1893
40		3413.83	-48.59	1463.40		0.0332
50		3413.83	-45.87	1465.25		0.0313
60		3413.83	-41.15	1465.25		0.0281
70		3413.83	-34.36	1465.25		0.0235



80		3413.83	-30.68	1465.25		0.0209
90		3413.83	-23.46	1465.25		0.0160
100		3413.83	-19.47	1465.25		0.0133
110		3413.83	-12.03	1465.25		0.0082
120	0.74	3413.83	-9.76	1462.78	0.0002	0.0067

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	96.96	1181.99	-394.29	1229.73	0.0820	0.3206
20	85.00	1181.99	-405.45	1194.74	0.0719	0.3394
30	57.56	905.46	-369.24	1068.80	0.0636	0.3455
40	726.01	1798.78		823.77	0.4036	
50	708.26	1798.78		827.03	0.3937	
60	704.64	1798.78		827.03	0.3917	
70	699.91	1798.78		827.03	0.3891	
80	698.34	1798.78		827.03	0.3882	
90	699.13	1798.78		827.03	0.3887	
100	698.51	1798.78		827.03	0.3883	
110	696.02	1798.78		827.03	0.3869	
120	719.98	1798.78		822.66	0.4003	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	619.44	1181.99	-907.59	1229.73	0.5241	0.7380
20	596.34	1181.99	-906.91	1194.74	0.5045	0.7591
30	479.36	905.46	-781.43	1068.80	0.5294	0.7311
40	828.23	1798.78		823.77	0.4604	
50	807.40	1798.78		827.03	0.4489	
60	782.39	1798.78		827.03	0.4350	
70	757.67	1798.78		827.03	0.4212	
80	737.65	1798.78		827.03	0.4101	
90	722.94	1798.78		827.03	0.4019	
100	710.57	1798.78		827.03	0.3950	
110	699.81	1798.78		827.03	0.3890	
120	720.72	1798.78		822.66	0.4007	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	96.96	1181.99	-394.29	1229.73	0.0820	0.3206
20	85.00	1181.99	-405.45	1194.74	0.0719	0.3394
30	57.56	905.46	-369.24	1068.80	0.0636	0.3455
40	0.92	1798.78	-103.92	1309.51	0.0005	0.0794
50	1.89	1798.78	-99.67	1313.38	0.0011	0.0759
60		1798.78	-87.77	1313.38		0.0668
70		1798.78	-74.17	1313.38		0.0565
80		1798.78	-63.92	1313.38		0.0487
90		1798.78	-51.08	1313.38		0.0389
100		1798.78	-41.60	1313.38		0.0317
110		1798.78	-29.81	1313.38		0.0227
120		1798.78	-25.91	1308.19		0.0198

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	619.44	1181.99	-907.59	1229.73	0.5241	0.7380
20	596.34	1181.99	-906.91	1194.74	0.5045	0.7591
30	479.36	905.46	-781.43	1068.80	0.5294	0.7311
40	103.15	1798.78	-203.59	1309.51	0.0573	0.1555
50	101.04	1798.78	-196.43	1313.38	0.0562	0.1496
60	76.03	1798.78	-163.57	1313.38	0.0423	0.1245
70	51.31	1798.78	-130.27	1313.38	0.0285	0.0992
80	31.29	1798.78	-101.84	1313.38	0.0174	0.0775



90	16.58	1798.78	-73.92	1313.38	0.0092	0.0563
100	4.21	1798.78	-52.97	1313.38	0.0023	0.0403
110		1798.78	-33.20	1313.38		0.0253
120		1798.78	-26.36	1308.19		0.0202

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-291.80	1229.73		0.2373
20		2004.25	-309.94	1194.74		0.2594
30		1706.92	-295.76	1375.58		0.2150
40	917.40	3413.83		1463.40	0.2687	
50	894.79	3413.83		1465.25	0.2621	
60	896.27	3413.83		1465.25	0.2625	
70	896.07	3413.83		1465.25	0.2625	
80	898.31	3413.83		1465.25	0.2631	
90	903.03	3413.83		1465.25	0.2645	
100	905.68	3413.83		1465.25	0.2653	
110	906.65	3413.83		1465.25	0.2656	
120	940.14	3413.83		1462.78	0.2754	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-243.43	1229.73		0.1980
20		2004.25	-264.75	1194.74		0.2216
30		1706.92	-260.37	1375.58		0.1893
40	908.92	3413.83		1463.40	0.2662	
50	886.57	3413.83		1465.25	0.2597	
60	890.13	3413.83		1465.25	0.2607	
70	891.69	3413.83		1465.25	0.2612	
80	895.38	3413.83		1465.25	0.2623	
90	901.28	3413.83		1465.25	0.2640	
100	904.83	3413.83		1465.25	0.2650	
110	906.40	3413.83		1465.25	0.2655	
120	940.10	3413.83		1462.78	0.2754	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-176.78	1229.73		0.1438
20		1181.99	-190.02	1194.74		0.1590
30		905.46	-185.64	1068.80		0.1737
40	689.54	1798.78		823.77	0.3833	
50	672.86	1798.78		827.03	0.3741	
60	676.42	1798.78		827.03	0.3760	
70	677.99	1798.78		827.03	0.3769	
80	681.67	1798.78		827.03	0.3790	
90	687.57	1798.78		827.03	0.3822	
100	691.13	1798.78		827.03	0.3842	
110	692.69	1798.78		827.03	0.3851	
120	718.76	1798.78		822.66	0.3996	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-134.63	1229.73		0.1095
20		1181.99	-143.61	1194.74		0.1202
30		905.46	-139.61	1068.80		0.1306
40	674.91	1798.78		823.77	0.3752	
50	658.73	1798.78		827.03	0.3662	
60	662.80	1798.78		827.03	0.3685	
70	667.16	1798.78		827.03	0.3709	
80	671.43	1798.78		827.03	0.3733	
90	678.14	1798.78		827.03	0.3770	



100	682.50	1798.78	827.03	0.3794
110	687.07	1798.78	827.03	0.3820
120	713.59	1798.78	822.66	0.3967

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-194.52	1229.73		0.1582
20		1181.99	-209.91	1194.74		0.1757
30		905.46	-205.53	1068.80		0.1923
40	687.51	1798.78		823.77	0.3822	
50	670.89	1798.78		827.03	0.3730	
60	674.45	1798.78		827.03	0.3749	
70	676.27	1798.78		827.03	0.3760	
80	680.61	1798.78		827.03	0.3784	
90	686.77	1798.78		827.03	0.3818	
100	690.98	1798.78		827.03	0.3841	
110	692.69	1798.78		827.03	0.3851	
120	718.76	1798.78		822.66	0.3996	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-176.78	1229.73		0.1438
20		1181.99	-190.02	1194.74		0.1590
30		905.46	-185.64	1068.80		0.1737
40	689.54	1798.78		823.77	0.3833	
50	672.86	1798.78		827.03	0.3741	
60	676.42	1798.78		827.03	0.3760	
70	677.99	1798.78		827.03	0.3769	
80	681.67	1798.78		827.03	0.3790	
90	687.57	1798.78		827.03	0.3822	
100	691.13	1798.78		827.03	0.3842	
110	692.69	1798.78		827.03	0.3851	
120	718.76	1798.78		822.66	0.3996	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-194.52	1229.73		0.1582
20		1181.99	-209.91	1194.74		0.1757
30		905.46	-205.53	1068.80		0.1923
40		1798.78	-50.62	823.77		0.0614
50		1798.78	-47.84	827.03		0.0579
60		1798.78	-43.12	827.03		0.0521
70		1798.78	-36.08	827.03		0.0436
80		1798.78	-31.74	827.03		0.0384
90		1798.78	-24.27	827.03		0.0293
100		1798.78	-19.62	827.03		0.0237
110		1798.78	-12.03	827.03		0.0145
120	0.74	1798.78	-9.76	822.66	0.0004	0.0119

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.83	984.99	-292.98	1024.78	0.0049	0.2859
20		984.99	-310.13	995.62		0.3115
30		754.55	-293.64	890.67		0.3297
40	710.32	1498.98		686.47	0.4739	
50	692.99	1498.98		689.19	0.4623	
60	693.33	1498.98		689.19	0.4625	
70	691.99	1498.98		689.19	0.4616	
80	693.18	1498.98		689.19	0.4624	
90	696.11	1498.98		689.19	0.4644	
100	697.17	1498.98		689.19	0.4651	



110	695.71	1498.98	689.19	0.4641
120	720.04	1498.98	685.55	0.4804

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.83	984.99	-292.98	1024.78	0.0049	0.2859
20		984.99	-310.13	995.62		0.3115
30		754.55	-293.64	890.67		0.3297
40		1498.98	-85.68	1091.26		0.0785
50		1498.98	-82.03	1094.49		0.0749
60		1498.98	-74.50	1094.49		0.0681
70		1498.98	-64.59	1094.49		0.0590
80		1498.98	-57.36	1094.49		0.0524
90		1498.98	-47.09	1094.49		0.0430
100		1498.98	-39.56	1094.49		0.0361
110		1498.98	-29.10	1094.49		0.0266
120		1498.98	-25.68	1090.16		0.0236

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-194.52	1229.73		0.1582
20		1181.99	-209.91	1194.74		0.1757
30		905.46	-205.53	1068.80		0.1923
40	687.51	1798.78		823.77	0.3822	
50	670.89	1798.78		827.03	0.3730	
60	674.45	1798.78		827.03	0.3749	
70	676.27	1798.78		827.03	0.3760	
80	680.61	1798.78		827.03	0.3784	
90	686.77	1798.78		827.03	0.3818	
100	690.98	1798.78		827.03	0.3841	
110	692.69	1798.78		827.03	0.3851	
120	718.76	1798.78		822.66	0.3996	

Absolute Maximum of the all of the Stress Ratio's 0.7591

Governing Element: K-2

Governing Load Case 4 : NP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
K-3	...	-144.1	48.2	715.3	619.4	+ Tens
K-3	...	-144.1	-48.2	-715.3	-907.6	- Comp
K-2	...	-155.3	52.3	699.3	596.3	+ Tens
K-2	...	-155.3	-52.3	-699.3	-906.9	- Comp
K-1	...	-151.0	52.3	578.1	479.4	+ Tens
K-1	...	-151.0	-52.3	-578.1	-781.4	- Comp
H-1	...	-36.8	13.2	140.1	116.6	+ Tens
H-1	...	-36.8	-13.2	-140.1	-190.2	- Comp
S-1	...	-34.6	12.7	136.0	114.1	+ Tens
S-1	...	-34.6	-12.7	-136.0	-183.3	- Comp
S-2	...	-30.7	12.2	107.6	89.1	+ Tens
S-2	...	-30.7	-12.2	-107.6	-150.5	- Comp
S-3	...	-26.4	9.5	81.2	64.4	+ Tens
S-3	...	-26.4	-9.5	-81.2	-117.2	- Comp
S-4	...	-22.2	9.5	57.0	44.4	+ Tens
S-4	...	-22.2	-9.5	-57.0	-88.7	- Comp



S-5	...	-15.6	8.9	36.4	29.7	+ Tens
S-5	...	-15.6	-8.9	-36.4	-60.8	- Comp
S-6	...	-11.3	8.7	19.9	17.3	+ Tens
S-6	...	-11.3	-8.7	-19.9	-39.9	- Comp
S-7	...	-6.8	5.7	7.6	6.5	+ Tens
S-7	...	-6.8	-5.7	-7.6	-20.1	- Comp
H-2	...	-5.1	5.3	2.5	2.7	+ Tens
H-2	...	-5.1	-5.3	-2.5	-12.8	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: K-1

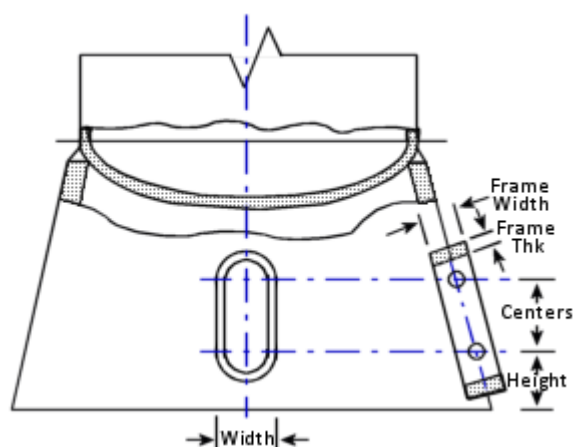
Total vessel (empty) weight	Fg	196253.703 kgf
Head Weight below section 2-2	DeltaFg	6788.939 kgf
Weight of vessel contents (liquid)	Ff	16646.326 kgf
Vessel Gross Weight	Fg + Ff	212900.031 kgf
Skirt Mean Diameter	Dz	2628.000 mm
Skirt Analysis Thickness (Corroded)	ez	16.400 mm
Skirt Height	H	600.000 mm
Additional Axial Force at section 1-1	F1	-3532.100 kgf
Moment at section 1-1	M1	489729.000 kgf-m.
Applied Moment - Bottom of Skirt	Mb	511090.781 kgf-m.
Weight at Bottom of Skirt	W	207476.688 kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	2809.500 mm
Head Corroded Thickness	eb	57.500 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 689 \text{ mm} \\
 K &= D_i / (2 * h_i) = 2752 / (2 * 689) = 2 \\
 R &= D_i (0.44 * K + 0.02) = 2752 * (0.44 * 2 + 0.02) = 2475 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 2752 * ((0.5 / 2) - 0.08) = 468 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	11689059.000 cm ⁴
Moment of Inertia	Iyy =	11689059.000 cm ⁴
Section Modulus	W =	88957904.000 mm ³
Section Metal Area	A =	1354.001 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

$$\begin{aligned}
 &\text{Radius of Gyration of Skirt at 4-4 [rg]:} \\
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(11689059, 11689059) / 1354)} \\
 &= 929 \text{ mm}
 \end{aligned}$$



Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned} &= r_g / L \\ &= 929/600 \\ &= 1.549 \end{aligned}$$

Correct the Moment [M4]:

top of skirt is the critical end
Controlling Moment [M4] = 489729 kgf-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$\begin{aligned} &= -M / Z - W / A \\ &= -489729/355831616 - 207477/1354 \\ &= -291 \text{ kgf/cm}^2 \end{aligned}$$

Upwind Side:

$$\begin{aligned} &= +(M) / Z - W / A \\ &= +489729/355831616 - 207477/1354 \\ &= -15.6 \text{ kgf/cm}^2 \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F_1 - F_g - F_f + 4 \cdot M_1 / D_z \\ &= -3532 - 196254 - 16646 + 4 \cdot 489729/2628 \\ &= 528969 \text{ kgf} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F_1 - F_g - F_f - 4 \cdot M_1 / D_z \\ &= -3532 - 196254 - 16646 - 4 \cdot 489729/2628 \\ &= -961833 \text{ kgf} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (F_{zp} + \Delta F_q) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (528969 + 6789 + 16646) / (\pi \cdot 2810 \cdot 57.5) + 5929 \cdot 2810 / (4 \cdot 57.5) \\ &= 847.44 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned} &= (F_{zq} + \Delta F_q + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (-961833 + 6789 + 16646) / (\pi \cdot 2810 \cdot 57.5) + 5929 \cdot 2810 / (4 \cdot 57.5) \\ &= 553.70 \text{ kgf/cm}^2 \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned} &= (F_f + \Delta F_q) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (16646 + 6789) / (\pi \cdot 2810 \cdot 57.5) + 5929 \cdot 2810 / (4 \cdot 57.5) \\ &= 743.22 \text{ kgf/cm}^2 \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned} &= F_{zp} / (\pi \cdot D_z \cdot e_z) \\ &= 528969 / (\pi \cdot 2628 \cdot 16.4) \\ &= 390.67 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned} &= F_{zq} / (\pi \cdot D_z \cdot e_z) \\ &= -961833 / (\pi \cdot 2628 \cdot 16.4) \\ &= -710.36 \text{ kgf/cm}^2 \end{aligned}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$



$$= 1 - 0.5(2810 + 57.5 - 2628 + 16.4) / (468 + 57.5) \\ = 0.76$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.76) = 40.8 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))} \\ = 0.5 \cdot \sqrt{(57.5^2 + 16.4^2 + 2 \cdot 57.5 \cdot 16.4 \cdot 0.76)} \\ = 35.4 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 35.36 \cdot 528968.8 = 18707.87 \text{ kgf-m.} \\ M_q = a \cdot F_{zq} = 35.36 \cdot -961833.0 = -34016.84 \text{ kgf-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2 \\ = 0.63 - 0.057(57.5/16.4)^2 \\ = -0.071$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ = -0.071 \cdot 6 \cdot 18708 / (\pi \cdot 2810 \cdot 57.5^2) \\ = -27.19 \text{ kgf/cm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ = -0.071 \cdot 6 \cdot -34017 / (\pi \cdot 2810 \cdot 57.5^2) \\ = 49.44 \text{ kgf/cm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ = -0.071 \cdot 6 \cdot 18708 / (\pi \cdot 2628 \cdot 16.4^2) \\ = -357.31 \text{ kgf/cm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ = -0.071 \cdot 6 \cdot -34017 / (\pi \cdot 2628 \cdot 16.4^2) \\ = 649.70 \text{ kgf/cm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 2.26/111 = 0.02 \\ y = 125 \cdot eb/D_b = 125 \cdot 57.5/2810 = 2.56$$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 2.56 = 3.69$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b (y / y_A) \cdot \alpha - 1 / (4 \cdot eb) \\ = (5929 + 0) \cdot 2810 (40.8/40) \cdot 3.69 - 1 / (4 \cdot 57.5) \\ = 2039.16 \text{ kgf/cm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ = 847 - -27.2 + 2039 \\ = 2913.79 \text{ kgf/cm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ = 847 + -27.2 - 2039 \\ = -1218.90 \text{ kgf/cm}^2$$



Stress [σ_{1qitot}]:
 $= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$
 $= 554 - 49.4 + 2039$
 $= 2543.42 \text{ kgf/cm}^2$

Stress [σ_{1qotot}]:
 $= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$
 $= 554 + 49.4 - 2039$
 $= -1436.03 \text{ kgf/cm}^2$

Stress [σ_{2pitot}]:
 $= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$
 $= 743 + -27.2 + 2039$
 $= 2755.19 \text{ kgf/cm}^2$

Stress [σ_{2potot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 743 - -27.2 - 2039$
 $= -1268.76 \text{ kgf/cm}^2$

Stress [σ_{2qitot}]:
 $= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$
 $= 743 + 49.4 + 2039$
 $= 2831.81 \text{ kgf/cm}^2$

Stress [σ_{2qotot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 743 - 49.4 - 2039$
 $= -1345.38 \text{ kgf/cm}^2$

Stress [σ_{3pitot}]:
 $= \sigma_{3pm} - \sigma_{3pb}$
 $= 391 - -357$
 $= 747.98 \text{ kgf/cm}^2$

Stress [σ_{3potot}]:
 $= \sigma_{3pm} + \sigma_{3pb}$
 $= 391 + -357$
 $= 33.37 \text{ kgf/cm}^2$

Stress [σ_{3qitot}]:
 $= \sigma_{3qm} - \sigma_{3qb}$
 $= -710 - 650$
 $= -1360.06 \text{ kgf/cm}^2$

Stress [σ_{3qotot}]:
 $= \sigma_{3qm} + \sigma_{3qb}$
 $= -710 + 650$
 $= -60.67 \text{ kgf/cm}^2$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$
 $= 1499 (3 - (1/1.5)) \cdot (847/1509)^2$
 $= 4181.81 \text{ kgf/cm}^2$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$
 $= 1499 (3 - (1/1.5)) \cdot (554/1509)^2$
 $= 4362.41 \text{ kgf/cm}^2$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$
 $= 1499 (3 - (1/1.5)) \cdot (743/1509)^2$
 $= 4254.56 \text{ kgf/cm}^2$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$



$$= 1499 (3 - (1/1.5) \cdot (743/1509)^2)$$
$$= 4254.56 \text{ kgf/cm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$
$$= 1499 (3 - (1/1.5) \cdot (391/1509)^2)$$
$$= 4429.97 \text{ kgf/cm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$
$$= 1499 (3 - (1/1.5) \cdot (-710/1509)^2)$$
$$= 4275.51 \text{ kgf/cm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (kgf/cm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	2913.79	4181.81	Pass	46
Stot1po	-1218.90	4181.81	Pass	47
Stot1qi	2543.42	4362.41	Pass	48
Stot1qo	-1436.03	4362.41	Pass	49
Stot2pi	2755.19	4254.56	Pass	50
Stot2po	-1268.76	4254.56	Pass	51
Stot2qi	2831.81	4254.56	Pass	52
Stot2qo	-1345.38	4254.56	Pass	53
Stot3pi	747.98	4429.97	Pass	54
Stot3po	33.37	4429.97	Pass	55
Stot3qi	-1360.06	4275.51	Pass	56
Stot3qo	-60.67	4275.51	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.

Packing is installed in the Field after being lifted.

Insulation is installed in the Shop.

Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	18091.664 mm
Center of Gravity of the Packing	16390.000 mm
Center of Gravity of the Liquid	7451.902 mm
Center of Gravity of the Insulation	13994.795 mm
Center of Gravity of the Lining	12109.214 mm
Center of Gravity of the Nozzles	15509.164 mm
Center of Gravity of the Added Weights (Operating)	17811.762 mm
Center of Gravity of the Added Weights (Empty)	17261.775 mm
Center of Gravity of Bare Shell New and Cold	14781.495 mm
Center of Gravity of Bare Shell Corroded	14837.413 mm
Vessel CG in the Operating Condition	15067.851 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	15407.375 mm
Vessel CG in the Test Condition	15657.691 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	2646.0000	mm
Skirt Thickness	STHK	20.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	44.0000	mm
Design Temperature of the Basing		82.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	1407.13	kgf/cm ²
Basing Yield Stress		2473.72	kgf/cm ²
Inside Diameter of Basing	DI, D4	2430.0000	mm
Outside Diameter of Basing	DOU, D5	3030.0000	mm

Nominal Diameter of Bolts	BND, d(b0)	64.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	1407.21	kgf/cm ²
Number of Bolts	RN, nb	20	
Diameter of Bolt Circle	DC, DBC	2837.9998	mm
Bolt Allowable Shear Stress		844.324	kgf/cm ²

Ultimate Comp. Strength of Concrete	FPC	421.8	kgf/cm ²
Allowable Comp. Strength of Concrete	FC	229.4	kgf/cm ²
Modular ratio Steel/Concrete		9.833	

Thickness of Gusset Plates	TGA, e7	20.0000	mm
Width of Gussets at Top Plate	TWDT	182.0000	mm
Width of Gussets at Base Plate	BWDT	182.0000	mm
Gusset Plate Elastic Modulus	E	2031470.0	kgf/cm ²
Gusset Plate Yield Stress	SY	2473.7	kgf/cm ²
Height of Gussets	HG, h1	312.0000	mm
Distance between Gussets	RG, b6	120.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	60.0000	mm
Number of Gussets per bolt	NG	2	

Thickness of Top Plate or Ring	TTA, e5	50.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	192.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	90.0000	mm

External Corrosion Allowance	CA	0.0000	mm
------------------------------	----	--------	----

Dead Weight of Vessel	DW	181418.2	kgf
Operating Weight of Vessel	ROW	225668.4	kgf
Earthquake Moment on Basing	EQMOM	648888.9	kgf-m.
Wind Moment on Basing	WIMOM	136362.2	kgf-m.

User Moment at the Base [Earthquake Case]		99682.2	kgf-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		99682.2	kgf-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	

Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	

Factor for Increase of Allowables	Fact	1.0000	
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Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 748571/2838) - 218702)/20$$

$$= 41818.3 \text{ kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 236044/2838) - 181418)/20$$

$$= 7563.6 \text{ kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/2838) - 225668)/20$$

$$= -11283.4 \text{ kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 22.1010 cm²
Area Available in all the Bolts Abss * RN : 442.0206 cm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.300	0.451	1.511	2.441	0.438	0.781	215150.8	433853.0	8082.8
4	0.376	0.338	1.705	2.277	0.421	0.784	219037.7	437739.8	8820.2
6	0.319	0.310	1.561	2.400	0.433	0.782	216079.8	434782.0	8255.8
8	0.315	0.312	1.548	2.410	0.434	0.782	215844.9	434547.0	8211.9
10	0.311	0.310	1.539	2.418	0.435	0.782	215669.8	434372.0	8179.3
12	0.310	0.310	1.536	2.420	0.435	0.782	215626.2	434328.4	8171.2

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 215626 / (4.96 \cdot 2838 \cdot 2.42)$$

$$= 1267 \text{ kgf/cm}^2, \text{ Should be } \leq 1407 \text{ kgf/cm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 20 \cdot 22.1 / (\pi \cdot 2838)$$

$$= 4.96 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 57.9[(2 \cdot 0.31 \cdot 2838 + 300) / (2 \cdot 0.31 \cdot 2838)]$$

$$= 67.8 \text{ kgf/cm}^2, \text{ Should be } \leq 229 \text{ kgf/cm}^2$$

Values for table 10.3, l = 192, b = 223, l/b = 0.86138833

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(2695, 4098) = 4098 \text{ kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 4098 / 1633)} + 0$$

$$= 38.8 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 192, b = 120, b/l = 0.625
As b/l (0.62) is less than 1, inverting b/l = 1.60.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 31101 / (4 \cdot \pi) [1.3(\ln((2 \cdot 192 \cdot \sin(\pi \cdot 96/192))) / (\pi \cdot 50))) + 1] - [(0.7 - 0.06) \cdot 31101 / (4 \cdot \pi)]$$

$$= 3766.2998 \text{ kgf}$$



Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{\frac{6 \cdot \text{Abs}(\text{Mo})}{F_b}} + C_a$$

$$= \sqrt{\frac{6 \cdot \text{Abs}(3766)}{1649}} + 0$$

$$= 37.0172 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot l_{avg}) + C_a$$

$$= 31101 / (1266 \cdot 182) + 0$$

$$= 13.5 \text{ (not less than } 9.52 + 0 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 2838 / 20 = 446 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 \left(\frac{P \cdot a}{m \cdot (h + t_{ba})} \cdot 1.5 \cdot S_{ktope} \right)^{2/3} \cdot r^{1/3} + C_a$$

$$= 1.76(31101 \cdot 96 / (446 \cdot 356 \cdot 2111))^{2/3} \cdot 13231^{1/3} + C_a$$

$$= 17.9 + 1.6 = 19.5 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 35904 / (22.1 \cdot 20)$$

$$= 81.2 \text{ kgf/cm}^2. \text{ Must be less than } 844.3 \text{ kgf/cm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	38.8071	mm
Actual Basing Thickness as entered by user	44.0000	mm
Required Thickness of Chair Cap	37.0172	mm
Actual Top Ring Thickness as entered by user	50.0000	mm
Required Gusset thickness, + CA	13.5030	mm
Actual Gusset Thickness as entered by user	20.0000	mm
Required Thickness of Skirt for Local Stress	19.4946	mm
Given Thickness of Skirt	20.0000	mm
Required Gusset Height to meet local stress	257.2916	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - C_A)$$

$$= 908 \cdot (20 - 1.6)$$

$$= 167 \text{ kgf/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 167 / [(0.4 \cdot 2474) \cdot 2 \cdot 0.707]$$

$$= 11.9 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cm}w_{th} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 31101 / (172 + 2 \cdot (312 + 50))$$

$$= 34.7 \text{ kgf/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cm}w_{th} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 31101 \cdot 96 / (172 \cdot (362) + 0.6667 \cdot (362)^2)$$

$$= 20 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{W_v^2 + W_h^2}$$

$$= \sqrt{34.7^2 + 20^2}$$

$$= 40 \text{ kgf/mm}$$



Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:
= $W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
= $40 / [(0.4 \cdot 2474) \cdot 2 \cdot 0.707]$
= 2.86 mm

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
= Bolt Load / (2 • TopWth)
= $31101 / (2 \cdot 182)$
= 85.4 kgf/mm

Weld Load [Wh]:
= Bolt Load • e / (2 • Hgt • TopWth)
= $31101 \cdot 96 / (2 \cdot 362 \cdot 182)$
= 22.7 kgf/mm

Resultant Weld Load [Wr]:
= $\sqrt{W_v^2 + W_h^2}$
= $\sqrt{85.4^2 + 22.7^2}$
= 88.4 kgf/mm

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
= $W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
= $88.4 / [(0.4 \cdot 2474) \cdot 2 \cdot 0.707]$
= 6.32 mm

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	11.9357	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	6.3178	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: D-10"

From: 40

Pressure for Reinforcement Calculations	P	6099.306	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Elliptical Head	D	2750.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	58.5000	mm
Head Internal Corrosion Allowance	c	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	55.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.5000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1875.0	mm
Centerline Distance to Tangent		1650.0	mm

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: D-10"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750	in.
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Actual Thickness Used in Calculation 0.520 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6099 \cdot 2752 \cdot 0.9 / (2 \cdot 1499 \cdot 1 - 0.2 \cdot 6099)$
 $= 51.5631$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6099 \cdot 137 / (1499 \cdot 1 + 0.4 \cdot 6099)$
 $= 5.5723$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.8541 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 66.8 / 0.5773$
 $= 115.6982$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	493.2934	mm
Parallel to Vessel Wall, opening length	d	246.6467	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	143.7500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1499/1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1499/1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	127.179	9.303	NA
Area in Shell	A1	14.643	123.215	NA
Area in Nozzle Wall	A2	21.934	35.499	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	168.799	168.799	NA
TOTAL AREA AVAILABLE	Atot	206.376	328.513	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (247 \cdot 51.6 \cdot 1 + 2 \cdot 80 \cdot 51.6 \cdot 1 \cdot (1 - 1))$$

$$= 127.179 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 247 (1 \cdot 57.5 - 1 \cdot 51.6) - 2 \cdot 80$$

$$(1 \cdot 57.5 - 1 \cdot 51.6) \cdot (1 - 1)$$

$$= 14.643 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 144) (13.2 - 5.57) 1$$

$$= 21.934 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(144, 250, 100)) \cdot (80 - 13.2) \cdot 1$$

$$= 168.799 \text{ cm}^2 \text{ Includes Bevel Area of } 35.202 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.5723 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 58.2767 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 58.2767 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.9672 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.1, \max(58.3, 2.5)]$$

$$= 8.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.57, 8.1)$$

$$= 8.1026 \text{ mm}$$

Available Nozzle Neck Thickness = 13.2016 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 566.8,	Allowable	: 1499.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3193.3 kgf/cm ²	Passed
Occasional	: 275.7,	Allowable	: 1993.6 kgf/cm ²	Passed
Shear	: 162.0,	Allowable	: 1049.3 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govern. thk, tg = 13.2, tr = 5.54, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.42$, Temp. Reduction = 46 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -18 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 6099.31/10340.00 = 0.590

Weld Size Calculations, Description: D-10"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (127 - 14.6 + 2 \cdot 80 \cdot 1 \cdot (1 \cdot 57.5 - 51.6)) \cdot 1499)$
 $= 182927.31 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (191 + 1 - 0 \cdot 1) \cdot 1499$
 $= 287403.44 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (191 + 0 + 1 + (92)) \cdot 1499$
 $= 425309.41 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (191 + 0 + 1 + 0 + (92)) \cdot 1499$
 $= 425309.41 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 273 \cdot 10 \cdot 0.49 \cdot 1499$
 $= 31503. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 130) \cdot (80 - 0) \cdot 0.7 \cdot 1499$
 $= 342628. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 273 \cdot (58.5 - 1) \cdot 0.74 \cdot 1499$



= 273563. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (31503 + 342628) = 374131 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (31503 + 0 + 273563 + 0) = 305066 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (31503 + 273563 + 0) = 305066 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 374131 kgf, must exceed W = 182927 kgf or W1 = 287403 kgf
 Path 2-2 = 305065 kgf, must exceed W = 182927 kgf or W2 = 425309 kgf
 Path 3-3 = 305065 kgf, must exceed W = 182927 kgf or W3 = 425309 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6091 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 3.6982 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 312.1982 mm

Input Echo, WRC107/537 Item 1, Description: D-10" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	4950.000	mm
Vessel Thickness	Tv	58.500	mm
Design Temperature	T1	430.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1498.98	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm
Nozzle Diameter	Dn	406.647	mm
Nozzle Thickness	Tn	80.000	mm
Nozzle Material		SA-336 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	1509.10	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1498.98	kgf/cm ²
Design Internal Pressure	Dp	6099.306	kPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

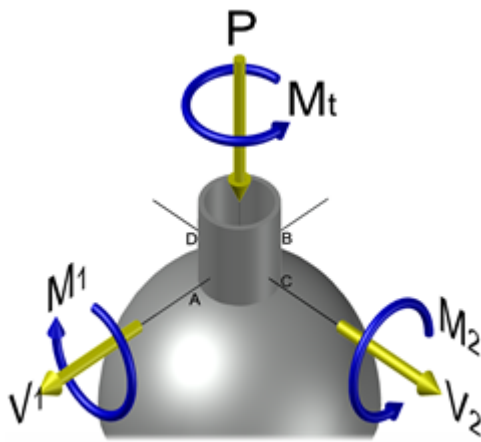
Radial Load (SUS)	P	2293.6	kgf
Longitudinal Shear (SUS)	(V1) V1	1656.5	kgf
Circumferential Shear (SUS)	(Vc) V2	1656.5	kgf
Circumferential Moment (SUS)	(Mc) M1	1274.3	kgf-m.

Longitudinal Moment	(SUS)	(Ml)	M2	1274.3	kgf-m.
Torsional Moment	(SUS)		Mt	1465.4	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 406.647 + 2 \cdot 1.65 \cdot \sqrt{(2504.75 (58.5 - 1.0))}$$

$$= 1659.009 \text{ mm}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load		P	2293.6	kgf
Circumferential Shear	(VC)	V2	1656.5	kgf
Longitudinal Shear	(VL)	V1	1656.5	kgf
Circumferential Moment	(MC)	M1	1274.3	kgf-m.
Longitudinal Moment	(ML)	M2	1274.3	kgf-m.
Torsional Moment		MT	1465.4	kgf-m.

Dimensionless Param: U = 0.54 TAU = 5.00 (2.04) RHO = 0.72

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.06995	(A,B,C,D)
$M(x) / P$	SP 1	0.08420	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / MC$	SM 1	0.10941	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / MC$	SM 1	0.20830	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / ML$	SM 1	0.10941	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / ML$	SM 1	0.20830	(A,B,C,D)
$N(y) * T / P$	SP 1	0.12239	(A,B,C,D)
$M(y) / P$	SP 1	0.02840	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / MC$	SM 1	0.09592	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / MC$	SM 1	0.08236	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / ML$	SM 1	0.09592	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / ML$	SM 1	0.08236	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-4.9	-4.9	-4.9	-4.9	-4.9	-4.9	-4.9	-4.9
Rad. Bend. P		-35.0	35.0	-35.0	35.0	-35.0	35.0	-35.0	35.0
Rad. Memb. MC		0.0	0.0	0.0	0.0	-11.1	-11.1	11.1	11.1
Rad. Bend. MC		0.0	0.0	0.0	0.0	-126.9	126.9	126.9	-126.9
Rad. Memb. ML		-11.1	-11.1	11.1	11.1	0.0	0.0	0.0	0.0
Rad. Bend. ML		-126.9	126.9	126.9	-126.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-177.9	146.0	98.1	-85.6	-177.9	146.0	98.1	-85.6
Tang. Memb. P		-8.5	-8.5	-8.5	-8.5	-8.5	-8.5	-8.5	-8.5
Tang. Bend. P		-11.8	11.8	-11.8	11.8	-11.8	11.8	-11.8	11.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-9.7	-9.7	9.7	9.7
Tang. Bend. MC		0.0	0.0	0.0	0.0	-50.2	50.2	50.2	-50.2
Tang. Memb. ML		-9.7	-9.7	9.7	9.7	0.0	0.0	0.0	0.0
Tang. Bend. ML		-50.2	50.2	50.2	-50.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-80.2	43.8	39.6	-37.1	-80.2	43.8	39.6	-37.1
Shear VC		4.5	4.5	-4.5	-4.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-4.5	-4.5	4.5	4.5
Shear MT		9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8
Tot. Shear		14.3	14.3	5.3	5.3	5.3	5.3	14.3	14.3
Str. Int.		180.0	148.0	98.6	86.2	178.2	146.3	101.5	89.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1339.6	1339.6	1339.6	1339.6	1339.6	1339.6	1339.6	1339.6
Rad. Pl (SUS)		-16.0	-16.0	6.3	6.3	-16.0	-16.0	6.3	6.3
Rad. Q (SUS)		-162.0	162.0	91.9	-91.9	-162.0	162.0	91.9	-91.9
Long. Pm (SUS)		1339.6	1339.6	1339.6	1339.6	1339.6	1339.6	1339.6	1339.6
Long. Pl (SUS)		-18.2	-18.2	1.3	1.3	-18.2	-18.2	1.3	1.3
Long. Q (SUS)		-62.0	62.0	38.4	-38.4	-62.0	62.0	38.4	-38.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.5	4.5	-4.5	-4.5	-4.5	-4.5	4.5	4.5
Shear Q (SUS)		9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8
Pm (SUS)		1339.6	1339.6	1339.6	1339.6	1339.6	1339.6	1339.6	1339.6
Pm+Pl (SUS)		1327.2	1327.2	1348.5	1348.5	1327.2	1327.2	1348.5	1348.5
Pm+Pl+Q (Total)		1261.4	1487.6	1438.2	1303.1	1259.7	1485.9	1441.1	1306.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1339.61	1498.98	Passed
Pm+Pl (SUS)	1348.52	2248.47	Passed



Pm+Pl+Q (TOTAL)	1487.59	4496.94	Passed
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*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: MH-3-24"

From: 50

Pressure for Reinforcement Calculations	P	6082.418	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1049.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		220.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-3-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 1.181 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6082 \cdot 1376 / (1499 \cdot 1 - 0.6 \cdot 6082)$
 $= 58.3857 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6082 \cdot 305 / (1499 \cdot 1 + 0.4 \cdot 6082)$
 $= 12.4067 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8178 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1099.2000	mm
Parallel to Vessel Wall, opening length	d	549.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		133.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	320.888	58.901	NA
Area in Shell	A1	3.376	206.461	NA
Area in Nozzle Wall	A2	46.798	74.964	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	16.674	16.674	NA
Area in Element	A5	255.199	255.199	NA
TOTAL AREA AVAILABLE	Atot	322.047	553.299	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness: 1047.6013 58.0000 mm
 Based on given Pad Diameter: 1049.6000 57.7365 mm
 Based on the Estimated Diameter Limit: 1097.6124 52.0562 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (550 \cdot 58.4 \cdot 1 + 2 \cdot 30 \cdot 58.4 \cdot 1 \cdot (1 - 1))$
 $= 320.888 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 550(1 \cdot 59 - 1 \cdot 58.4) - 2 \cdot 30$
 $(1 \cdot 59 - 1 \cdot 58.4) \cdot (1 - 1)$
 $= 3.376 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 133) \cdot (30 - 12.4) \cdot 1$
 $= 46.798 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$
 $= (1) \cdot 1 + (0) \cdot 1 + 1567^2 \cdot 1$
 $= 16.674 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (1050 - 610) \cdot 58 \cdot 1$
 $= 255.199 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 12.4067 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= ta$
 $= 12.4067 \text{ mm}$

Available Nozzle Neck Thickness = 30.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)

Govrn. thk, $tg = 30$, $tr = 12.3$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, $tg = 30$, $tr = 12.3$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, $tg = 30$, $tr = 12.3$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C

Shell to Pad Weld Junction at Pad OD, Curve: A



Govrn. thk, tg = 58, c = 1 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.98$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	39 °C
Min Metal Temp. at Required thickness (UCS 66.1)	39 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 30, tr = 12.3, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C

Gov. MDMT of the Nozzle	:	-24 °C
Gov. MDMT of the Reinforcement Pad	:	39 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	39 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6082.42 / 10340.00 = 0.588$

Weld Size Calculations, Description: MH-3-24"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$31.1080 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (321 - 3.38 + 2 \cdot 30 \cdot 1 \cdot (1 \cdot 59 - 58.4)) \cdot 1499)$
 $= 476495.84 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (46.8 + 255 + 16.7 - 0 \cdot 1) \cdot 1499$
 $= 477680.91 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (46.8 + 0 + 1 + (35.4)) \cdot 1499$
 $= 124711.95 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (46.8 + 0 + 16.7 + 255 + (35.4)) \cdot 1499$
 $= 530744.69 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 1499$
 $= 70332. \text{ kgf}$



Shear, Pad Element Weld [Spew]:
= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
= $(3.14/2.0) \cdot 1050 \cdot 44 \cdot 0.49 \cdot 1499$
= 532828. kgf

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.14 \cdot 290) \cdot (30 - 0) \cdot 0.7 \cdot 1499$
= 286591. kgf

Tension, Pad Groove Weld [Tpgw]:
= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
= $(3.14/2) \cdot 610 \cdot 58 \cdot 0.74 \cdot 1499$
= 616054. kgf

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
= $(3.14/2.0) \cdot 610 \cdot (58 - 0) \cdot 0.74 \cdot 1499$
= 616055. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= $(SPEW + SNW) = (532828 + 286591) = 819418 \text{ kgf}$

Failure Path [Path 2-2]:
= $(Sonw + Tpgw + Tngw + Sinw)$
= $(70332 + 616054 + 616055 + 0) = 1302441 \text{ kgf}$

Failure Path [Path 3-3]:
= $(Spew + Tngw + Sinw)$
= $(532828 + 616055 + 0) = 1148883 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 819418 kgf, must exceed W = 476495 kgf or W1 = 477680 kgf
Path 2-2 = 1302441 kgf, must exceed W = 476495 kgf or W2 = 124711 kgf
Path 3-3 = 1148883 kgf, must exceed W = 476495 kgf or W3 = 530744 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6093 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 34.2085 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 594.2085 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 5.176 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

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Input, Nozzle Desc: E-8"

From: 50

Pressure for Reinforcement Calculations	P	6089.965	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1009.30	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	819.0750	mm
Thickness of Pad	te	64.0000	mm
Weld leg size between Pad and Shell	Wp	60.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	60.0000	mm
Reinforcing Pad Width		300.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: E-8"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 8.625 in.
 Actual Thickness Used in Calculation 0.793 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6090 \cdot 1376 / (1499 \cdot 1 - 0.6 \cdot 6090)$
 $= 58.4600$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6090 \cdot 110 / (1009 \cdot 1 + 0.4 \cdot 6090)$
 $= 6.5779$ mm

Required Nozzle thickness under External Pressure per UG-28 : 1.0023 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	357.6066	mm
Parallel to Vessel Wall, opening length	d	178.8033	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		114.3396	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1009 / 1499)$
 $= 0.673$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1009 / 1499)$
 $= 0.673$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.67, 1)$
 $= 0.673$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	112.219	20.572	NA
Area in Shell	A1	0.894	62.227	NA
Area in Nozzle Wall	A2	20.876	29.461	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.693	2.693	NA
Area in Element	A5	88.660	88.660	NA
TOTAL AREA AVAILABLE	Atot	113.124	183.041	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 356.1935 64.0000 mm
 Based on given Pad Diameter: 819.0750 63.3472 mm
 Based on the Estimated Diameter Limit: 356.0191 64.0815 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (179 \cdot 58.5 \cdot 1 + 2 \cdot 20.1 \cdot 58.5 \cdot 1 \cdot (1 - 0.67))$
 $= 112.219 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 179(1 \cdot 59 - 1 \cdot 58.5) - 2 \cdot 20.1$
 $(1 \cdot 59 - 1 \cdot 58.5) \cdot (1 - 0.67)$
 $= 0.894 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 114) \cdot (20.1 - 6.58) \cdot 0.67$
 $= 20.876 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 20^2 \cdot 0.67 + (0)^2 \cdot 0.67 + 0^2 \cdot 1$
 $= 2.693 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (358 - 219) \cdot 64 \cdot 1$
 $= 88.660 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.5779 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 59.4600 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 59.4600 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.9685 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 7.1600 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.16, \max(59.5, 2.5)]$
 $= 7.1600 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(6.58, 7.16)$
 $= 7.1600 \text{ mm}$

Available Nozzle Neck Thickness = 20.1358 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 339.4,	Allowable	: 1009.3 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2426.3 kgf/cm ²	Passed
Occasional	: 123.9,	Allowable	: 1342.4 kgf/cm ²	Passed
Shear	: 125.9,	Allowable	: 706.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B



Govrn. thk, tg = 20.1, tr = 5.54, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 20.1, tr = 5.54, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 20.1, tr = 5.54, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 13 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 60, tr = 58.1, c = 1 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 39 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 20.1, tr = 5.54, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : 39 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 39 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6089.97 / 10340.00 = 0.589$

Weld Size Calculations, Description: E-8"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$13.3000 = 0.7 * t_{min}$	$14.1400 = 0.7 * W_o$ mm
Pad Weld	$9.5000 = 0.5 * t_{minPad}$	$42.4200 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 * t_n * f_r1 * (E1 * t - tr)) S_v)$
 $= \max(0, (112 - 0.89 + 2 * 20.1 * 0.67 * (1 * 59 - 58.5)) 1499)$
 $= 167092.98 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
$$= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$
$$= (20.9 + 88.7 + 2.69 - 0 \cdot 0.67) \cdot 1499$$
$$= 168229.12 \text{ kgf}$$

Weld Load [W2]:
$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$
$$= (20.9 + 0 + 2.69 + (16)) \cdot 1499$$
$$= 59310.93 \text{ kgf}$$

Weld Load [W3]:
$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$
$$= (20.9 + 0 + 2.69 + 88.7 + (16)) \cdot 1499$$
$$= 192210.36 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$
$$= (3.14/2.0) \cdot 219 \cdot 20 \cdot 0.49 \cdot 1009$$
$$= 34038. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:
$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$
$$= (3.14/2.0) \cdot 819 \cdot 60 \cdot 0.49 \cdot 1499$$
$$= 567003. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:
$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.14 \cdot 99.5) \cdot (20.1 - 0) \cdot 0.7 \cdot 1009$$
$$= 44456. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:
$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$
$$= (3.14/2) \cdot 219 \cdot 60 \cdot 0.74 \cdot 1499$$
$$= 229029. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:
$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$
$$= (3.14/2.0) \cdot 219 \cdot (60 - 1) \cdot 0.74 \cdot 1499$$
$$= 225212. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
$$= (SPEW + SNW) = (567003 + 44456) = 611459 \text{ kgf}$$

Failure Path [Path 2-2]:
$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (34038 + 229029 + 225212 + 0) = 488278 \text{ kgf}$$

Failure Path [Path 3-3]:
$$= (Spew + Tngw + Sinw)$$
$$= (567003 + 225212 + 0) = 792215 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 611458 kgf, must exceed W = 167092 kgf or W1 = 168229 kgf
Path 2-2 = 488278 kgf, must exceed W = 167092 kgf or W2 = 59310 kgf
Path 3-3 = 792214 kgf, must exceed W = 167092 kgf or W3 = 192210 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6115 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 4.3700 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 564.3700 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 178.80 mm

Input Echo, WRC297 Item 1, Description: E-8" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2750.000	mm	
Shell Thickness	60.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 11		
Vessel Cold S.I. Allowable	Smc 1509.10	kgf/cm ²	
Vessel Hot S.I. Allowable	Smh 1498.98	kgf/cm ²	
Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	0.0000	mm	
Nozzle Diameter	219.075	mm	
Nozzle Thickness	20.1358	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-335 P11		
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²	
Nozzle Hot S.I. Allowable	SNmh 1009.30	kgf/cm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	819.075	mm
Thickness of Reinforcing Pad	64.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	6089.97	kPa
Radial Load	P	1834.88	kgf
Circumferential Shear	Vc	1325.19	kgf
Longitudinal Shear	Vl	1325.19	kgf
Circumferential Moment	Mc	815.53	kgf-m.
Longitudinal Moment	Ml	815.53	kgf-m.
Torsional Moment	Mt	937.86	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (3.651) is < 5
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(1437.5(60.0 - 1.0))}$
 $= 1180.120 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2875.000	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	123.000	mm
Nozzle Thickness used	(t) =	20.136	mm
T / t	=	6.109	
d / t	=	10.880	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.368$$

Nr/P = 0.180
Mr/P = 0.111
MØ/P = 0.269
NØ/P = 0.270
MrD/Mc = 0.111
NrDL/Mc = 0.153
MØd/Mc = 0.301
NØDL/Mc = -0.118
MrD/Ml = 0.110
NrDL/Ml = 0.155
MØD/Ml = 0.293
NØDL/Ml = 0.081

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-19	19	-19	19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-43	43	43	-43
Normal Pressure Stress	664	727	664	727
Outplane Stress Summary	596	784	687	702

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-8	8	-8	8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-16	16	16	-16
Inplane Pressure Stress	332	332	332	332
Inplane Stress Summary	302	350	342	325

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	10	10	10	10
Shear Stress Summary	13	13	6	6

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	597	784	687	702

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-19	19	-19	19

Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	332	332	332	332

Outplane Stress Summary	268	396	351	301
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-8	8	-8	8
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-16	16	16	-16
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	664	727	664	727

Inplane Stress Summary	634	745	674	720
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	10	10	10	10

Shear Stress Summary	6	6	13	13
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	634	745	675	720
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	297	371	297	371

Hoop Stress Summary	292	366	296	370
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	-14	-14	-14	-14
Axial Bending (P)	-57	57	-57	57
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-142	-142	142	142
Axial Bending (ML)	-180	180	180	-180
Axial Pressure Stress	148	148	148	148



Axial Stress Summary -245 229 398 153

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	19	19	-19	-19
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	61	61	61	61
Shear Stress Summary	80	80	42	42

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	561	404	414	378

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	297	371	297	371
Hoop Stress Summary	296	371	291	365

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-14	-14	-14	-14
Axial Bending (P)	-57	57	-57	57
Axial Membrane (Mc)	-142	-142	142	142
Axial Bending (Mc)	-188	188	188	-188
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	148	148	148	148
Axial Stress Summary	-254	238	407	144

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-19	-19	19	19
Shear due to Torsion	61	61	61	61
Shear Stress Summary	42	42	80	80

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	557	383	448	392

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 • 1.65 • $\sqrt{(R_{mean}(t - ca))}$



$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{1405.5(60.0 - 1.0)}$$

$$= 1169.363 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2811.000 mm
Nozzle Outside Diameter	(d) =	819.075 mm
Vessel Thickness used	(T) =	59.000 mm
Nozzle Thickness used	(t) =	320.136 mm
T / t	=	0.184
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 2.011$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.072
Mr/P	=	0.145
M0/P	=	0.043
N0/P	=	0.075
MrD/Mc	=	0.289
NrDL/Mc	=	0.093
M0d/Mc	=	0.088
N0DL/Mc	=	0.075
MrD/M1	=	0.110
NrDL/M1	=	0.091
M0D/M1	=	0.034
N0DL/M1	=	0.098

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-5	5	5	-5
Normal Pressure Stress	1417	1480	1417	1480
Outplane Stress Summary	1391	1492	1409	1486

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-45	45	-45	45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-18	18	18	-18
Inplane Pressure Stress	708	708	708	708
Inplane Stress Summary	637	767	680	734

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	3	3	0	0

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1391	1492	1409	1486

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-15	15	15	-15
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	708	708	708	708
Outplane Stress Summary	674	731	708	705

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-45	45	-45	45
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-49	49	49	-49
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1417	1480	1417	1480
Inplane Stress Summary	1316	1569	1420	1475

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	3	3

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1316	1569	1420	1475

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	664	727	664	727	664	727	664	727
Circ. Pl (SUS)	-4	-4	-2	-2	-5	-5	1	1
Circ. Q (SUS)	-62	62	24	-24	-24	24	8	-8

Long. Pm (SUS)	332	332	332	332	332	332	332	332
Long. Pl (SUS)	-5	-5	1	1	-1	-1	-5	-5
Long. Q (SUS)	-24	24	8	-8	-63	63	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	664.0	727.0	664.0	727.0	664.0	727.0	664.0	727.0
Pm+Pl (SUS)	660.0	723.0	662.0	725.0	659.0	722.0	665.0	728.0
Pm+Pl+Q (Total)	598.6	785.4	686.1	701.1	635.1	746.1	673.5	720.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	727.00	1498.98	Passed
Pm+Pl (SUS)	728.02	2248.47	Passed
Pm+Pl+Q (TOTAL)	785.39	4496.94	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1417	1480	1417	1480	1417	1480	1417	1480
Circ. Pl (SUS)	-5	-5	-1	-1	-5	-5	-1	-1
Circ. Q (SUS)	-18	18	-8	8	-94	94	4	-4
Long. Pm (SUS)	708	708	708	708	708	708	708	708
Long. Pl (SUS)	-5	-5	-1	-1	-5	-5	-1	-1
Long. Q (SUS)	-63	63	-27	27	-28	28	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	1417	1480	1417	1480	1417	1480	1417	1480
Pm+Pl (SUS)	1412	1475	1416	1479	1412	1475	1416	1479
Pm+Pl+Q (Total)	1394	1493	1408	1487	1318	1569	1420	1475

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1480.00	1498.98	Passed
Pm+Pl (SUS)	1479.00	2248.47	Passed
Pm+Pl+Q (TOTAL)	1569.00	4496.94	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int.	Stress Values at
---------------------	------------------

Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	297	371	297	371	297	371	297	371
Circ. Pl (SUS)	-4	-4	-2	-2	-1	-1	-5	-5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	148	148	148	148	148	148	148	148
Long. Pl (SUS)	-156	-156	128	128	-156	-156	128	128
Long. Q (SUS)	-237	237	123	-123	-245	245	131	-131
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	19	19	-19	-19	-19	-19	19	19
Shear Q (SUS)	61	61	61	61	61	61	61	61
Pm (SUS)	297.0	371.0	297.0	371.0	297.0	371.0	297.0	371.0
Pm+Pl (SUS)	303.4	376.9	306.7	372.7	306.4	379.9	304.6	369.8
Pm+Pl+Q (Total)	561.3	403.6	413.8	376.9	555.4	382.2	448.0	391.9

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	371.00	1009.30	Passed
Pm+Pl (SUS)	379.91	1513.95	Passed
Pm+Pl+Q (TOTAL)	561.29	3027.90	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: UC-2-14"

From: 60

Pressure for Reinforcement Calculations	P	6069.965	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5250.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	655.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		150.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: UC-2-14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 14.000 in.
 Actual Thickness Used in Calculation 0.709 in.

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6070 \cdot 1376 / (1499 \cdot 1 - 0.6 \cdot 6070)$
 $= 58.2632 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6070 \cdot 178 / (1499 \cdot 1 + 0.4 \cdot 6070)$
 $= 7.2227 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3308 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	639.2000	mm
Parallel to Vessel Wall, opening length	d	319.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		103.0000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	186.209	34.252	NA
Area in Shell	A1	2.355	120.060	NA
Area in Nozzle Wall	A2	22.201	34.338	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	164.488	164.488	NA
TOTAL AREA AVAILABLE	Atot	190.044	319.886	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	632.5882	58.0000 mm
Based on given Pad Diameter:	655.5999	56.6478 mm
Based on the Estimated Diameter Limit:	637.6125	56.9667 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (320 \cdot 58.3 \cdot 1 + 2 \cdot 18 \cdot 58.3 \cdot 1 \cdot (1 - 1))$
 $= 186.209 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 320(1 \cdot 59 - 1 \cdot 58.3) - 2 \cdot 18$
 $(1 \cdot 59 - 1 \cdot 58.3) \cdot (1 - 1)$
 $= 2.355 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 103) \cdot (18 - 7.22) \cdot 1$
 $= 22.201 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (639 - 356) \cdot 58 \cdot 1$
 $= 164.488 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 7.2227 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= ta$
 $= 7.2227 \text{ mm}$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)

Govrn. thk, $tg = 18$, $tr = 7.17$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.4$, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, $tg = 18$, $tr = 7.17$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.4$, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, $tg = 18$, $tr = 7.17$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.4$, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C



Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 58, c = 1 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.98$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	39 °C
Min Metal Temp. at Required thickness (UCS 66.1)	38 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 18, tr = 7.17, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.4$, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Gov. MDMT of the Nozzle	:	-42 °C
Gov. MDMT of the Reinforcement Pad	:	38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	38 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6069.96 / 10340.00 = 0.587$

Weld Size Calculations, Description: UC-2-14"

Intermediate Calc. for nozzle/shell Welds	Tmin	18.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$31.1080 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (186 - 2.35 + 2 \cdot 18 \cdot 1 \cdot (1 \cdot 59 - 58.3)) \cdot 1499)$
 $= 275990.88 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (22.2 + 164 + 1 - 0 \cdot 1) \cdot 1499$
 $= 281341.53 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (22.2 + 0 + 1 + (21.2)) \cdot 1499$
 $= 66616.40 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (22.2 + 0 + 1 + 164 + (21.2)) \cdot 1499$
 $= 313179.75 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 356 \cdot 10 \cdot 0.49 \cdot 1499$

= 41027. kgf

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.14/2.0) \cdot 656 \cdot 44 \cdot 0.49 \cdot 1499 \\ &= 332814. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 169) \cdot (18 - 0) \cdot 0.7 \cdot 1499 \\ &= 100158. \text{ kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.14/2) \cdot 356 \cdot 58 \cdot 0.74 \cdot 1499 \\ &= 359365. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 356 \cdot (58 - 0) \cdot 0.74 \cdot 1499 \\ &= 359366. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (332814 + 100158) = 432973 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (41027 + 359365 + 359366 + 0) = 759758 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ &= (332814 + 359366 + 0) = 692180 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 432972 kgf, must exceed W = 275990 kgf	or W1 = 281341 kgf
Path 2-2 = 759757 kgf, must exceed W = 275990 kgf	or W2 = 66616 kgf
Path 3-3 = 692179 kgf, must exceed W = 275990 kgf	or W3 = 313179 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6129 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 11.5441 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 571.5441 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 5.332 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

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Input, Nozzle Desc: B-14"

From: 60

Pressure for Reinforcement Calculations	P	6069.965	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3650.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	655.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		150.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B-14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 14.000 in.
 Actual Thickness Used in Calculation 0.709 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6070 \cdot 1376 / (1499 \cdot 1 - 0.6 \cdot 6070)$
 $= 58.2632 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6070 \cdot 178 / (1499 \cdot 1 + 0.4 \cdot 6070)$
 $= 7.2227 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3308 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	639.2000	mm
Parallel to Vessel Wall, opening length	d	319.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		103.0000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	186.209	34.252	NA
Area in Shell	A1	2.355	120.060	NA
Area in Nozzle Wall	A2	22.201	34.338	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	164.488	164.488	NA
TOTAL AREA AVAILABLE	Atot	190.044	319.886	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.



SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 632.5882 58.0000 mm
 Based on given Pad Diameter: 655.5999 56.6478 mm
 Based on the Estimated Diameter Limit: 637.6125 56.9667 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (320 \cdot 58.3 \cdot 1 + 2 \cdot 18 \cdot 58.3 \cdot 1 \cdot (1 - 1))$
 $= 186.209 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 320(1 \cdot 59 - 1 \cdot 58.3) - 2 \cdot 18$
 $(1 \cdot 59 - 1 \cdot 58.3) \cdot (1 - 1)$
 $= 2.355 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 103) \cdot (18 - 7.22) \cdot 1$
 $= 22.201 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (639 - 356) \cdot 58 \cdot 1$
 $= 164.488 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 7.2227 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 59.2632 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 59.2632 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 1.9685 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.33, \max(59.3, 2.5)]$
 $= 8.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(7.22, 8.33)$
 $= 8.3312 \text{ mm}$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 507.2,	Allowable	: 1499.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3252.9 kgf/cm ²	Passed
Occasional	: 260.1,	Allowable	: 1993.6 kgf/cm ²	Passed
Shear	: 135.7,	Allowable	: 1049.3 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)



Govrn. thk, tg = 18, tr = 7.17, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.4$, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, tg = 18, tr = 7.17, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.4$, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 18, tr = 7.17, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.4$, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 58, c = 1 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.98$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A 39 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 38 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 18, tr = 7.17, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.4$, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C

Gov. MDMT of the Nozzle : -42 °C
 Gov. MDMT of the Reinforcement Pad : 38 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 38 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6069.96 / 10340.00 = 0.587$

Weld Size Calculations, Description: B-14"

Intermediate Calc. for nozzle/shell Welds Tmin 18.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	31.1080 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)$
 $= \max(0, (186 - 2.35 + 2 * 18 * 1 * (1 * 59 - 58.3))1499)$
 $= 275990.88 \text{ kgf}$



Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= $(A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
= $(22.2 + 164 + 1 - 0 \cdot 1) \cdot 1499$
= 281341.53 kgf

Weld Load [W2]:
= $(A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
= $(22.2 + 0 + 1 + (21.2)) \cdot 1499$
= 66616.40 kgf

Weld Load [W3]:
= $(A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
= $(22.2 + 0 + 1 + 164 + (21.2)) \cdot 1499$
= 313179.75 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= $(\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
= $(3.14/2.0) \cdot 356 \cdot 10 \cdot 0.49 \cdot 1499$
= 41027. kgf

Shear, Pad Element Weld [Spew]:
= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
= $(3.14/2.0) \cdot 656 \cdot 44 \cdot 0.49 \cdot 1499$
= 332814. kgf

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.14 \cdot 169) \cdot (18 - 0) \cdot 0.7 \cdot 1499$
= 100158. kgf

Tension, Pad Groove Weld [Tpgw]:
= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
= $(3.14/2) \cdot 356 \cdot 58 \cdot 0.74 \cdot 1499$
= 359365. kgf

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
= $(3.14/2.0) \cdot 356 \cdot (58 - 0) \cdot 0.74 \cdot 1499$
= 359366. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= $(SPEW + SNW) = (332814 + 100158) = 432973 \text{ kgf}$

Failure Path [Path 2-2]:
= $(Sonw + Tpgw + Tngw + Sinw)$
= $(41027 + 359365 + 359366 + 0) = 759758 \text{ kgf}$

Failure Path [Path 3-3]:
= $(Spew + Tngw + Sinw)$
= $(332814 + 359366 + 0) = 692180 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 432972 kgf, must exceed W = 275990 kgf or W1 = 281341 kgf
Path 2-2 = 759757 kgf, must exceed W = 275990 kgf or W2 = 66616 kgf
Path 3-3 = 692179 kgf, must exceed W = 275990 kgf or W3 = 313179 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6129 kPa

Nozzle is O.K. for the External Pressure: 103 kPa



The Drop for this Nozzle is : 11.5441 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 571.5441 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 5.332 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Input Echo, WRC297 Item 1, Description: B-14" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2750.000	mm	
Shell Thickness	60.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 11		
Vessel Cold S.I. Allowable	Smc 1509.10	kgf/cm ²	
Vessel Hot S.I. Allowable	Smh 1498.98	kgf/cm ²	
Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	0.0000	mm	
Nozzle Diameter	355.600	mm	
Nozzle Thickness	18.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-387 11		
Nozzle Cold S.I. Allowable	SNmc 1509.10	kgf/cm ²	
Nozzle Hot S.I. Allowable	SNmh 1498.98	kgf/cm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	655.600	mm
Thickness of Reinforcing Pad	58.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	6069.96	kPa
Radial Load	P	3211.03	kgf
Circumferential Shear	Vc	2319.08	kgf
Longitudinal Shear	Vl	2319.08	kgf
Circumferential Moment	Mc	2497.55	kgf-m.
Longitudinal Moment	Ml	2497.55	kgf-m.
Torsional Moment	Mt	2872.18	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{mean}(t - ca))}$
 $= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(1434.5(60.0 - 1.0))}$
 $= 1315.642 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2869.000	mm
Nozzle Outside Diameter	(d) =	355.600	mm
Vessel Thickness used	(T) =	117.000	mm
Nozzle Thickness used	(t) =	18.000	mm
T / t	=	6.500	
d / t	=	19.756	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.614$$

Nr/P = 0.093
Mr/P = 0.060
M0/P = 0.214
N0/P = 0.349
MrD/Mc = 0.095
NrDL/Mc = 0.106
M0d/Mc = 0.307
N0DL/Mc = 0.124
MrD/Ml = 0.094
NrDL/Ml = 0.104
M0D/Ml = 0.288
N0DL/Ml = 0.216

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-8	-8	-8	-8
Outplane Bending (P)	-30	30	-30	30
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-11	-11	11	11
Outplane Bending (ML)	-88	88	88	-88
Normal Pressure Stress	698	760	698	760
Outplane Stress Summary	560	859	759	704

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-8	8	-8	8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-29	29	29	-29
Inplane Pressure Stress	349	349	349	349
Inplane Stress Summary	304	379	372	331

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	12	12	12	12
Shear Stress Summary	15	15	8	8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	561	860	759	704

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-8	-8	-8	-8
Outplane Bending (P)	-30	30	-30	30
Outplane Membrane (Mc)	-6	-6	6	6

Outplane Bending (Mc)	-94	94	94	-94
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	349	349	349	349

Outplane Stress Summary	210	459	411	282
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-8	8	-8	8
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-29	29	29	-29
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	698	760	698	760

Inplane Stress Summary	653	790	722	742
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	12	12	12	12

Shear Stress Summary	8	8	15	15
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	653	790	723	743
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	-8	-8	-8	-8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	624	698	624	698

Hoop Stress Summary	605	679	627	701
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	-16	-16	-16	-16
Axial Bending (P)	-80	80	-80	80
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-162	-162	162	162
Axial Bending (ML)	-550	550	550	-550
Axial Pressure Stress	312	312	312	312

Axial Stress Summary	-498	763	928	-12
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	23	23	-23	-23
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	80	80	80	80
Shear Stress Summary	103	103	57	57

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1122	833	939	723

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	-8	-8	-8	-8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	624	698	624	698
Hoop Stress Summary	609	684	622	696

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	-16	-16	-16	-16
Axial Bending (P)	-80	80	-80	80
Axial Membrane (Mc)	-162	-162	162	162
Axial Bending (Mc)	-543	543	543	-543
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	312	312	312	312
Axial Stress Summary	-490	755	920	-4

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	D1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-23	-23	23	23
Shear due to Torsion	80	80	80	80
Shear Stress Summary	57	57	103	103

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	D1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1106	787	953	731

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(1405.5(60.0 - 1.0))}$$

= 1305.888 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2811.000 mm
Nozzle Outside Diameter (d) = 655.600 mm
Vessel Thickness used (T) = 59.000 mm
Nozzle Thickness used (t) = 168.000 mm
T / t = 0.351
d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.610$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.079
Mr/P = 0.167
M0/P = 0.049
N0/P = 0.099
MrD/Mc = 0.295
NrDL/Mc = 0.092
M0d/Mc = 0.091
N0DL/Mc = 0.065
MrD/Ml = 0.135
NrDL/Ml = 0.084
M0D/Ml = 0.043
N0DL/Ml = 0.120

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-9	-9	-9	-9
Outplane Bending (P)	-26	26	-26	26
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	1413	1475	1413	1475
Outplane Stress Summary	1336	1507	1418	1477

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-7	-7	-7	-7
Inplane Bending (P)	-92	92	-92	92
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-88	88	88	-88
Inplane Pressure Stress	706	706	706	706
Inplane Stress Summary	509	871	704	712

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	7	7	7	7
Shear Stress Summary	10	10	3	3

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	698	760	698	760	698	760	698	760
Circ. Pl (SUS)	-19	-19	3	3	-7	-7	3	3
Circ. Q (SUS)	-118	118	58	-58	-37	37	21	-21
Long. Pm (SUS)	349	349	349	349	349	349	349	349



Long. P1 (SUS)	-7	-7	3	3	-14	-14	-2	-2
Long. Q (SUS)	-37	37	21	-21	-124	124	64	-64
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	698.0	760.0	698.0	760.0	698.0	760.0	698.0	760.0
Pm+P1 (SUS)	679.0	741.0	701.0	763.0	691.0	753.0	701.0	763.0
Pm+P1+Q (Total)	561.9	859.5	759.2	705.2	654.2	790.2	722.7	742.5

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	760.00	1498.98	Passed
Pm+P1 (SUS)	763.02	2248.47	Passed
Pm+P1+Q (TOTAL)	859.47	4496.94	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1413	1475	1413	1475	1413	1475	1413	1475
Circ. P1 (SUS)	-22	-22	4	4	-16	-16	2	2
Circ. Q (SUS)	-53	53	1	-1	-285	285	101	-101
Long. Pm (SUS)	706	706	706	706	706	706	706	706
Long. P1 (SUS)	-16	-16	2	2	-16	-16	-2	-2
Long. Q (SUS)	-180	180	-4	4	-86	86	34	-34
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	1413	1475	1413	1475	1413	1475	1413	1475
Pm+P1 (SUS)	1391	1453	1417	1479	1397	1459	1415	1477
Pm+P1+Q (Total)	1338	1506	1418	1478	1112	1744	1516	1376

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1475.00	1498.98	Passed
Pm+P1 (SUS)	1479.01	2248.47	Passed
Pm+P1+Q (TOTAL)	1744.02	4496.94	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	624	698	624	698	624	698	624	698
Circ. Pl (SUS)	-19	-19	3	3	-14	-14	-2	-2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	312	312	312	312	312	312	312	312
Long. Pl (SUS)	-178	-178	146	146	-178	-178	146	146
Long. Q (SUS)	-630	630	470	-470	-623	623	463	-463
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	23	23	-23	-23	-23	-23	23	23
Shear Q (SUS)	80	80	80	80	80	80	80	80
Pm (SUS)	624.0	698.0	624.0	698.0	624.0	698.0	624.0	698.0
Pm+Pl (SUS)	606.1	680.0	630.1	703.2	611.1	685.0	625.2	698.2
Pm+Pl+Q (Total)	1120	832	938	722	1104	788	953	730

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	698.00	1498.98	Passed
Pm+Pl (SUS)	703.16	2248.47	Passed
Pm+Pl+Q (TOTAL)	1120.11	4496.93	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH-2-24"

From: 80

Pressure for Reinforcement Calculations	P	6069.965	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		10550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1049.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		220.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-2-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 1.181 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6070 \cdot 1376 / (1499 \cdot 1 - 0.6 \cdot 6070)$
 $= 58.2632 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6070 \cdot 305 / (1499 \cdot 1 + 0.4 \cdot 6070)$
 $= 12.3817 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8178 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1099.2000	mm
Parallel to Vessel Wall, opening length	d	549.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		133.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	320.214	58.901	NA
Area in Shell	A1	4.050	206.461	NA
Area in Nozzle Wall	A2	46.865	74.964	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	16.674	16.674	NA
Area in Element	A5	255.199	255.199	NA
TOTAL AREA AVAILABLE	Atot	322.787	553.299	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness:	1045.1639	58.0000	mm
Based on given Pad Diameter:	1049.6000	57.4152	mm
Based on the Estimated Diameter Limit:	1097.6124	51.7665	mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (550 \cdot 58.3 \cdot 1 + 2 \cdot 30 \cdot 58.3 \cdot 1 \cdot (1 - 1))$
 $= 320.214 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 550(1 \cdot 59 - 1 \cdot 58.3) - 2 \cdot 30$
 $(1 \cdot 59 - 1 \cdot 58.3) \cdot (1 - 1)$
 $= 4.050 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 133) \cdot (30 - 12.4) \cdot 1$
 $= 46.865 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$
 $= (1) \cdot 1 + (0) \cdot 1 + 1567^2 \cdot 1$
 $= 16.674 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (1050 - 610) \cdot 58 \cdot 1$
 $= 255.199 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 12.3817 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= ta$
 $= 12.3817 \text{ mm}$

Available Nozzle Neck Thickness = 30.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)

Govrn. thk, $tg = 30$, $tr = 12.3$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, $tg = 30$, $tr = 12.3$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, $tg = 30$, $tr = 12.3$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 58, c = 1 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.98$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	39 °C
Min Metal Temp. at Required thickness (UCS 66.1)	38 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 30, tr = 12.3, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C

Gov. MDMT of the Nozzle	:	-24 °C
Gov. MDMT of the Reinforcement Pad	:	38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	38 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6069.96/10340.00 = 0.587$

Weld Size Calculations, Description: MH-2-24"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot Wo \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot TminPad$	$31.1080 = 0.7 \cdot Wp \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (320 - 4.05 + 2 \cdot 30 \cdot 1 \cdot (1 \cdot 59 - 58.3)) \cdot 1499)$
 $= 474586.47 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (46.9 + 255 + 16.7 - 0 \cdot 1) \cdot 1499$
 $= 477780.59 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (46.9 + 0 + 1 + (35.4)) \cdot 1499$
 $= 124811.58 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (46.9 + 0 + 16.7 + 255 + (35.4)) \cdot 1499$
 $= 530844.25 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 1499$
 $= 70332. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
= $(3.14/2.0) \cdot 1050 \cdot 44 \cdot 0.49 \cdot 1499$
= 532828. kgf

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.14 \cdot 290) \cdot (30 - 0) \cdot 0.7 \cdot 1499$
= 286591. kgf

Tension, Pad Groove Weld [Tpgw]:
= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
= $(3.14/2) \cdot 610 \cdot 58 \cdot 0.74 \cdot 1499$
= 616054. kgf

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
= $(3.14/2.0) \cdot 610 \cdot (58 - 0) \cdot 0.74 \cdot 1499$
= 616055. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= $(SPEW + SNW) = (532828 + 286591) = 819418 \text{ kgf}$

Failure Path [Path 2-2]:
= $(Sonw + Tpgw + Tngw + Sinw)$
= $(70332 + 616054 + 616055 + 0) = 1302441 \text{ kgf}$

Failure Path [Path 3-3]:
= $(Spew + Tngw + Sinw)$
= $(532828 + 616055 + 0) = 1148883 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 819418 kgf, must exceed W = 474586 kgf or W1 = 477780 kgf
Path 2-2 = 1302441 kgf, must exceed W = 474586 kgf or W2 = 124811 kgf
Path 3-3 = 1148883 kgf, must exceed W = 474586 kgf or W3 = 530844 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6093 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 34.2085 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 594.2085 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 5.176 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

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Input, Nozzle Desc: UC-1-14"

From: 90

Pressure for Reinforcement Calculations	P	6069.965	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	655.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		150.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: UC-1-14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 14.000 in.
 Actual Thickness Used in Calculation 0.709 in.

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6070 \cdot 1376 / (1499 \cdot 1 - 0.6 \cdot 6070)$
 $= 58.2632 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6070 \cdot 178 / (1499 \cdot 1 + 0.4 \cdot 6070)$
 $= 7.2227 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3308 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	639.2000	mm
Parallel to Vessel Wall, opening length	d	319.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		103.0000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	186.209	34.252	NA
Area in Shell	A1	2.355	120.060	NA
Area in Nozzle Wall	A2	22.201	34.338	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	164.488	164.488	NA
TOTAL AREA AVAILABLE	Atot	190.044	319.886	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	632.5882	58.0000 mm
Based on given Pad Diameter:	655.5999	56.6478 mm
Based on the Estimated Diameter Limit:	637.6125	56.9667 mm

Area Required [A]:
= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)
= (320•58.3•1+2•18•58.3•1•(1-1))
= 186.209 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 320 (1 • 59 - 1 • 58.3) - 2 • 18
(1 • 59 - 1 • 58.3) • (1 - 1)
= 2.355 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:
= (2 • Tlwp) • (tn - trn) • fr2
= (2 • 103) • (18 - 7.22) • 1
= 22.201 cm²

Area Available in Welds [A41 + A42 + A43]:
= Wo² • fr3 + (Wi-can/0.707)² • fr2 + Wp² • fr4
= 10² • 1 + (0)² • 1 + 0² • 1
= 1.000 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (639 - 356) • 58 • 1
= 164.488 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures ta = 7.2227 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= ta
= 7.2227 mm

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)

Govrn. thk, tg = 18, tr = 7.17, c = 0 mm, E* = 1
Thickness Ratio = tr • E*/(tg - c) = 0.4, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, tg = 18, tr = 7.17, c = 0 mm, E* = 1
Thickness Ratio = tr • E*/(tg - c) = 0.4, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 18, tr = 7.17, c = 0 mm, E* = 1
Thickness Ratio = tr • E*/(tg - c) = 0.4, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C



Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 58, c = 1 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.98$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	39 °C
Min Metal Temp. at Required thickness (UCS 66.1)	38 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 18, tr = 7.17, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.4$, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Gov. MDMT of the Nozzle	:	-42 °C
Gov. MDMT of the Reinforcement Pad	:	38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	38 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6069.96 / 10340.00 = 0.587$

Weld Size Calculations, Description: UC-1-14"

Intermediate Calc. for nozzle/shell Welds	Tmin	18.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{\text{minPad}}$	$31.1080 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (186 - 2.35 + 2 \cdot 18 \cdot 1 \cdot (1 \cdot 59 - 58.3)) \cdot 1499)$
 $= 275990.88 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - C_{an} / 707)^2 \cdot f_{r2}) \cdot S_v$
 $= (22.2 + 164 + 1 - 0 \cdot 1) \cdot 1499$
 $= 281341.53 \text{ kgf}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (22.2 + 0 + 1 + (21.2)) \cdot 1499$
 $= 66616.40 \text{ kgf}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (22.2 + 0 + 1 + 164 + (21.2)) \cdot 1499$
 $= 313179.75 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 356 \cdot 10 \cdot 0.49 \cdot 1499$



= 41027. kgf

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.14/2.0) \cdot 656 \cdot 44 \cdot 0.49 \cdot 1499 \\ &= 332814. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 169) \cdot (18 - 0) \cdot 0.7 \cdot 1499 \\ &= 100158. \text{ kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.14/2) \cdot 356 \cdot 58 \cdot 0.74 \cdot 1499 \\ &= 359365. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 356 \cdot (58 - 0) \cdot 0.74 \cdot 1499 \\ &= 359366. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (332814 + 100158) = 432973 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (41027 + 359365 + 359366 + 0) = 759758 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ &= (332814 + 359366 + 0) = 692180 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 432972 kgf, must exceed W = 275990 kgf	or W1 = 281341 kgf
Path 2-2 = 759757 kgf, must exceed W = 275990 kgf	or W2 = 66616 kgf
Path 3-3 = 692179 kgf, must exceed W = 275990 kgf	or W3 = 313179 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6129 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 11.5441 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 571.5441 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 5.332 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

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Input, Nozzle Desc: A-10"

From: 100

Pressure for Reinforcement Calculations	P	6069.965	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		16900.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1009.30	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	593.0500	mm
Thickness of Pad	te	62.0000	mm
Weld leg size between Pad and Shell	Wp	46.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	60.0000	mm
Reinforcing Pad Width		160.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A-10"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 10.750 in.
Actual Thickness Used in Calculation 0.984 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6070 \cdot 1376 / (1499 \cdot 1 - 0.6 \cdot 6070)$
 $= 58.2632 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6070 \cdot 137 / (1009 \cdot 1 + 0.4 \cdot 6070)$
 $= 8.1723 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1401 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	446.0875	mm
Parallel to Vessel Wall, opening length	d	223.0437	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		124.5078	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1009 / 1499)$
 $= 0.673$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1009 / 1499)$
 $= 0.673$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.67, 1)$
 $= 0.673$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	139.470	25.655	NA
Area in Shell	A1	1.523	77.651	NA
Area in Nozzle Wall	A2	28.220	40.011	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		2.693	2.693	NA
Area in Element	A5	107.283	107.283	NA
TOTAL AREA AVAILABLE	Atot	139.719	227.639	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 445.6852 62.0000 mm
 Based on given Pad Diameter: 593.0500 61.8559 mm
 Based on the Estimated Diameter Limit: 444.5000 62.4286 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (223 \cdot 58.3 \cdot 1 + 2 \cdot 25 \cdot 58.3 \cdot 1 \cdot (1 - 0.67))$
 $= 139.470 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 223(1 \cdot 59 - 1 \cdot 58.3) - 2 \cdot 25$
 $(1 \cdot 59 - 1 \cdot 58.3) \cdot (1 - 0.67)$
 $= 1.523 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 125) \cdot (25 - 8.17) \cdot 0.67$
 $= 28.220 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-can}/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 20^2 \cdot 0.67 + (0)^2 \cdot 0.67 + 0^2 \cdot 1$
 $= 2.693 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (446 - 273) \cdot 62 \cdot 1$
 $= 107.283 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 8.1723 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 59.2632 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 59.2632 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 1.9685 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 8.1026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.1, \max(59.3, 2.5)]$
 $= 8.1026 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(8.17, 8.1)$
 $= 8.1723 \text{ mm}$

Available Nozzle Neck Thickness = 25.0031 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 298.4,	Allowable	: 1009.3 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2467.2 kgf/cm ²	Passed
Occasional	: 124.1,	Allowable	: 1342.4 kgf/cm ²	Passed
Shear	: 101.7,	Allowable	: 706.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B



Govrn. thk, tg = 25, tr = 6.88, c = 0 mm, E* = 1
Thickness Ratio = $tr * E* / (tg - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 25, tr = 6.88, c = 0 mm, E* = 1
Thickness Ratio = $tr * E* / (tg - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 25, tr = 6.88, c = 0 mm, E* = 1
Thickness Ratio = $tr * E* / (tg - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 19 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 60, tr = 57.9, c = 1 mm, E* = 1
Thickness Ratio = $tr * E* / (tg - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 40 °C
Min Metal Temp. at Required thickness (UCS 66.1) 39 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 25, tr = 6.88, c = 0 mm, E* = 1
Thickness Ratio = $tr * E* / (tg - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : 39 °C
Gov. MDMT of the nozzle to shell joint welded assembly : 39 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $6069.96 / 10340.00 = 0.587$

Weld Size Calculations, Description: A-10"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$13.3000 = 0.7 * t_{min}$	$14.1400 = 0.7 * W_o$ mm
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$32.5220 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= $\max(0, (A - A1 + 2 * t_n * f_r1 * (E1 * t - tr)) S_v)$
= $\max(0, (139 - 1.52 + 2 * 25 * 0.67 * (1 * 59 - 58.3)) 1499)$
= 207151.33 kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (28.2 + 107 + 2.69 - 0 \cdot 0.67) \cdot 1499 \\ &= 207153.25 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (28.2 + 0 + 2.69 + (19.9)) \cdot 1499 \\ &= 76116.40 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (28.2 + 0 + 2.69 + 107 + (19.9)) \cdot 1499 \\ &= 236931.28 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 273 \cdot 20 \cdot 0.49 \cdot 1009 \\ &= 42424. \text{ kgf} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.14/2.0) \cdot 593 \cdot 46 \cdot 0.49 \cdot 1499 \\ &= 314745. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 124) \cdot (25 - 0) \cdot 0.7 \cdot 1009 \\ &= 68828. \text{ kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.14/2) \cdot 273 \cdot 60 \cdot 0.74 \cdot 1499 \\ &= 285456. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 273 \cdot (60 - 1) \cdot 0.74 \cdot 1499 \\ &= 280699. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (314745 + 68828) = 383574 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (42424 + 285456 + 280699 + 0) = 608579 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (314745 + 280699 + 0) = 595444 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 383573 kgf, must exceed W = 207151 kgf	or W1 = 207153 kgf
Path 2-2 = 608578 kgf, must exceed W = 207151 kgf	or W2 = 76116 kgf
Path 3-3 = 595444 kgf, must exceed W = 207151 kgf	or W3 = 236931 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6075 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 6.7946 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 566.7947 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 191.90 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 223.04 mm

Input Echo, WRC297 Item 1, Description: A-10" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2750.000	mm	
Shell Thickness	60.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 11		
Vessel Cold S.I. Allowable	Smc 1509.10	kgf/cm ²	
Vessel Hot S.I. Allowable	Smh 1498.98	kgf/cm ²	
Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	0.0000	mm	
Nozzle Diameter	273.050	mm	
Nozzle Thickness	25.0031	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-335 P11		
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²	
Nozzle Hot S.I. Allowable	SNmh 1009.30	kgf/cm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	593.050	mm
Thickness of Reinforcing Pad	62.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	6069.96	kPa
Radial Load	P	2293.59	kgf
Circumferential Shear	Vc	1656.48	kgf
Longitudinal Shear	Vl	1656.48	kgf
Circumferential Moment	Mc	1274.26	kgf-m.
Longitudinal Moment	Ml	1274.26	kgf-m.
Torsional Moment	Mt	1465.41	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (4.551) is < 5
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1436.5(60.0 - 1.0))}$
 $= 1233.761 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2873.000	mm
Nozzle Outside Diameter	(d) =	273.050	mm
Vessel Thickness used	(T) =	121.000	mm
Nozzle Thickness used	(t) =	25.003	mm
T / t	=	4.839	
d / t	=	10.921	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.463$$

Nr/P = 0.180
Mr/P = 0.116
MØ/P = 0.257
NØ/P = 0.263
MrD/Mc = 0.117
NrDL/Mc = 0.156
MØd/Mc = 0.293
NØDL/Mc = -0.108
MrD/Ml = 0.116
NrDL/Ml = 0.158
MØD/Ml = 0.285
NØDL/Ml = 0.149

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-24	24	-24	24
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-54	54	54	-54
Normal Pressure Stress	674	736	674	736
Outplane Stress Summary	586	805	705	706

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-10	10	-10	10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	337	337	337	337
Inplane Stress Summary	296	362	350	327

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	10	10	10	10
Shear Stress Summary	13	13	7	7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	587	806	705	706

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-24	24	-24	24

Outplane Membrane (Mc)	3	3	-3	-3
Outplane Bending (Mc)	-56	56	56	-56
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	337	337	337	337

Outplane Stress Summary	256	416	361	297
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-10	10	-10	10
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-22	22	22	-22
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	674	736	674	736

Inplane Stress Summary	633	761	687	726
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	10	10	10	10

Shear Stress Summary	7	7	13	13
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	633	761	688	727
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	297	372	297	372

Hoop Stress Summary	289	363	298	372
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	-11	-11	-11	-11
Axial Bending (P)	-55	55	-55	55
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-114	-114	114	114
Axial Bending (ML)	-167	167	167	-167
Axial Pressure Stress	148	148	148	148



Axial Stress Summary -201 245 363 140

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	15	15	-15	-15
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	49	49	49	49

Shear Stress Summary 65 65 34 34

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	507	392	378	377

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	3	3	-3	-3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	297	372	297	372

Hoop Stress Summary 297 371 290 364

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-11	-11	-11	-11
Axial Bending (P)	-55	55	-55	55
Axial Membrane (Mc)	-114	-114	114	114
Axial Bending (Mc)	-176	176	176	-176
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	148	148	148	148

Axial Stress Summary -210 254 373 131

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-15	-15	15	15
Shear due to Torsion	49	49	49	49

Shear Stress Summary 34 34 65 65

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	512	380	409	381

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 • 1.65 • $\sqrt{(Rmean(t - ca))}$



$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1405.5(60.0 - 1.0))}$$

$$= 1223.338 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2811.000 mm
Nozzle Outside Diameter	(d) =	593.050 mm
Vessel Thickness used	(T) =	59.000 mm
Nozzle Thickness used	(t) =	185.003 mm
T / t	=	0.319
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.456$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.085
Mr/P	=	0.179
M0/P	=	0.052
N0/P	=	0.103
MrD/Mc	=	0.300
NrDL/Mc	=	0.088
M0d/Mc	=	0.092
N0DL/Mc	=	0.061
MrD/M1	=	0.147
NrDL/M1	=	0.089
M0D/M1	=	0.046
N0DL/M1	=	0.120

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-20	20	-20	20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	1413	1475	1413	1475
Outplane Stress Summary	1361	1498	1410	1479

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-70	70	-70	70
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-54	54	54	-54
Inplane Pressure Stress	706	706	706	706
Inplane Stress Summary	570	820	690	722

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	7	7	1	1

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1361	1498	1410	1479

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-20	20	-20	20
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-34	34	34	-34
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	706	706	706	706
Outplane Stress Summary	641	750	717	689

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-70	70	-70	70
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-111	111	111	-111
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1413	1475	1413	1475
Inplane Stress Summary	1220	1646	1453	1434

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	1	1	7	7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1220	1646	1453	1434

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	674	736	674	736	674	736	674	736
Circ. Pl (SUS)	-8	-8	0	0	-6	-6	2	2
Circ. Q (SUS)	-78	78	30	-30	-32	32	12	-12



Long. Pm (SUS)	337	337	337	337	337	337	337	337
Long. Pl (SUS)	-6	-6	2	2	-1	-1	-7	-7
Long. Q (SUS)	-32	32	12	-12	-80	80	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	674.0	736.0	674.0	736.0	674.0	736.0	674.0	736.0
Pm+Pl (SUS)	666.0	728.0	674.0	736.0	668.0	730.0	676.0	738.0
Pm+Pl+Q (Total)	588.6	806.4	704.1	706.1	636.1	762.1	688.5	726.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	736.00	1498.98	Passed
Pm+Pl (SUS)	738.02	2248.47	Passed
Pm+Pl+Q (TOTAL)	806.38	4496.94	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1413	1475	1413	1475	1413	1475	1413	1475
Circ. Pl (SUS)	-13	-13	1	1	-10	-10	0	0
Circ. Q (SUS)	-37	37	-3	3	-181	181	41	-41
Long. Pm (SUS)	706	706	706	706	706	706	706	706
Long. Pl (SUS)	-10	-10	0	0	-9	-9	-3	-3
Long. Q (SUS)	-124	124	-16	16	-54	54	14	-14
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	1413	1475	1413	1475	1413	1475	1413	1475
Pm+Pl (SUS)	1400	1462	1414	1476	1403	1465	1413	1475
Pm+Pl+Q (Total)	1363	1499	1411	1479	1222	1646	1454	1434

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1475.00	1498.98	Passed
Pm+Pl (SUS)	1476.01	2248.47	Passed
Pm+Pl+Q (TOTAL)	1646.00	4496.94	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int.	Stress Values at
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Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	297	372	297	372	297	372	297	372
Circ. Pl (SUS)	-8	-8	0	0	-1	-1	-7	-7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	148	148	148	148	148	148	148	148
Long. Pl (SUS)	-125	-125	103	103	-125	-125	103	103
Long. Q (SUS)	-222	222	112	-112	-231	231	121	-121
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	15	15	-15	-15	-15	-15	15	15
Shear Q (SUS)	49	49	49	49	49	49	49	49
Pm (SUS)	297.0	372.0	297.0	372.0	297.0	372.0	297.0	372.0
Pm+Pl (SUS)	289.8	364.7	301.5	373.8	296.8	371.6	295.1	366.9
Pm+Pl+Q (Total)	504.5	391.9	377.4	376.9	508.6	380.2	407.0	381.3

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	372.00	1009.30	Passed
Pm+Pl (SUS)	373.83	1513.95	Passed
Pm+Pl+Q (TOTAL)	508.57	3027.90	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH-1-24"

From: 110

Pressure for Reinforcement Calculations	P	6069.965	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		18910.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm²
Diameter of Pad along vessel surface	Dp	1049.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		220.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 1.181 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6070 \cdot 1376 / (1499 \cdot 1 - 0.6 \cdot 6070)$
 $= 58.2632 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6070 \cdot 305 / (1499 \cdot 1 + 0.4 \cdot 6070)$
 $= 12.3817 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8178 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1099.2000	mm
Parallel to Vessel Wall, opening length	d	549.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		133.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499 / 1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	320.214	58.901	NA
Area in Shell	A1	4.050	206.461	NA
Area in Nozzle Wall	A2	46.865	74.964	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	16.674	16.674	NA
Area in Element	A5	255.199	255.199	NA
TOTAL AREA AVAILABLE	Atot	322.787	553.299	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness: 1045.1639 58.0000 mm
 Based on given Pad Diameter: 1049.6000 57.4152 mm
 Based on the Estimated Diameter Limit: 1097.6124 51.7665 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (550 \cdot 58.3 \cdot 1 + 2 \cdot 30 \cdot 58.3 \cdot 1 \cdot (1 - 1))$
 $= 320.214 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 550(1 \cdot 59 - 1 \cdot 58.3) - 2 \cdot 30$
 $(1 \cdot 59 - 1 \cdot 58.3) \cdot (1 - 1)$
 $= 4.050 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 133) \cdot (30 - 12.4) \cdot 1$
 $= 46.865 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$
 $= (1) \cdot 1 + (0) \cdot 1 + 1567^2 \cdot 1$
 $= 16.674 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (1050 - 610) \cdot 58 \cdot 1$
 $= 255.199 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 12.3817 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= ta$
 $= 12.3817 \text{ mm}$

Available Nozzle Neck Thickness = 30.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)

Govrn. thk, $tg = 30$, $tr = 12.3$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, $tg = 30$, $tr = 12.3$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, $tg = 30$, $tr = 12.3$, $c = 0 \text{ mm}$, $E* = 1$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C

Shell to Pad Weld Junction at Pad OD, Curve: A



Govrn. thk, tg = 58, c = 1 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.98$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	39 °C
Min Metal Temp. at Required thickness (UCS 66.1)	38 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 30, tr = 12.3, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C

Gov. MDMT of the Nozzle	:	-24 °C
Gov. MDMT of the Reinforcement Pad	:	38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	38 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6069.96/10340.00 = 0.587$

Weld Size Calculations, Description: MH-1-24"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot Wo \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot TminPad$	$31.1080 = 0.7 \cdot Wp \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (320 - 4.05 + 2 \cdot 30 \cdot 1 \cdot (1 \cdot 59 - 58.3)) \cdot 1499)$
 $= 474586.47 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (46.9 + 255 + 16.7 - 0 \cdot 1) \cdot 1499$
 $= 477780.59 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (46.9 + 0 + 1 + (35.4)) \cdot 1499$
 $= 124811.58 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (46.9 + 0 + 16.7 + 255 + (35.4)) \cdot 1499$
 $= 530844.25 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 1499$
 $= 70332. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
= $(3.14/2.0) \cdot 1050 \cdot 44 \cdot 0.49 \cdot 1499$
= 532828. kgf

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.14 \cdot 290) \cdot (30 - 0) \cdot 0.7 \cdot 1499$
= 286591. kgf

Tension, Pad Groove Weld [Tpgw]:
= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
= $(3.14/2) \cdot 610 \cdot 58 \cdot 0.74 \cdot 1499$
= 616054. kgf

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
= $(3.14/2.0) \cdot 610 \cdot (58 - 0) \cdot 0.74 \cdot 1499$
= 616055. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= $(SPEW + SNW) = (532828 + 286591) = 819418 \text{ kgf}$

Failure Path [Path 2-2]:
= $(Sonw + Tpgw + Tngw + Sinw)$
= $(70332 + 616054 + 616055 + 0) = 1302441 \text{ kgf}$

Failure Path [Path 3-3]:
= $(Spew + Tngw + Sinw)$
= $(532828 + 616055 + 0) = 1148883 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 819418 kgf, must exceed W = 474586 kgf or W1 = 477780 kgf
Path 2-2 = 1302441 kgf, must exceed W = 474586 kgf or W2 = 124811 kgf
Path 3-3 = 1148883 kgf, must exceed W = 474586 kgf or W3 = 530844 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6093 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 34.2085 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 594.2085 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 5.176 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

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Input, Nozzle Desc: C-14"

From: 120

Pressure for Reinforcement Calculations	P	6069.965	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Elliptical Head	D	2750.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	58.0000	mm
Head Internal Corrosion Allowance	c	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	655.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		150.0000	mm
Flange Class		900	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: C-14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 14.000 in.
 Actual Thickness Used in Calculation 0.630 in.

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6070 \cdot 2752 \cdot 0.9 / (2 \cdot 1499 \cdot 1 - 0.2 \cdot 6070)$
 $= 51.3141 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6070 \cdot 178 / (1499 \cdot 1 + 0.4 \cdot 6070)$
 $= 7.2227 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3308 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	647.2000	mm
Parallel to Vessel Wall, opening length	d	323.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		98.0000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1499/1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1499/1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 1499/1499)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	166.052	12.202	NA
Area in Shell	A1	18.400	160.049	NA
Area in Nozzle Wall	A2	17.204	28.752	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	169.128	169.128	NA
TOTAL AREA AVAILABLE	Atot	205.731	358.928	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 578.7887 58.0000 mm
 Based on given Pad Diameter: 655.6000 44.3928 mm
 Based on the Estimated Diameter Limit: 645.6125 44.6358 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (324 \cdot 51.3 \cdot 1 + 2 \cdot 16 \cdot 51.3 \cdot 1 \cdot (1 - 1))$
 $= 166.052 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 324(1 \cdot 57 - 1 \cdot 51.3) - 2 \cdot 16$
 $(1 \cdot 57 - 1 \cdot 51.3) \cdot (1 - 1)$
 $= 18.400 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 98) \cdot (16 - 7.22) \cdot 1$
 $= 17.204 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (647 - 356) \cdot 58 \cdot 1$
 $= 169.128 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.2227 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 58.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 58.0000 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.9672 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.33, \max(58, 2.5)]$
 $= 8.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(7.22, 8.33)$
 $= 8.3312 \text{ mm}$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 571.7,	Allowable	: 1499.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3188.4 kgf/cm ²	Passed
Occasional	: 298.2,	Allowable	: 1993.6 kgf/cm ²	Passed
Shear	: 149.5,	Allowable	: 1049.3 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)



Govrn. thk, tg = 16, tr = 7.17, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.45$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, tg = 16, tr = 7.17, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.45$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 16, tr = 7.17, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.45$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 58, tr = 51, c = 1 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.89$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 39 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 34 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 16, tr = 7.17, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.45$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C

Gov. MDMT of the Nozzle : -34 °C
 Gov. MDMT of the Reinforcement Pad : 34 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 34 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6069.96 / 15510.00 = 0.391$

Weld Size Calculations, Description: C-14"

Intermediate Calc. for nozzle/shell Welds Tmin 16.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * W_o$ mm
 Pad Weld 9.5000 = $0.5 * T_{minPad}$ 31.1080 = $0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_r \cdot (E_1 \cdot t - tr)) S_v)$
 $= \max(0, (166 - 18.4 + 2 \cdot 16 \cdot 1 \cdot (1 \cdot 57 - 51.3)) 1499)$
 $= 224055.53 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (17.2 + 169 + 1 - 0 \cdot 1) \cdot 1499$
 $= 280805.28 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (17.2 + 0 + 1 + (18.2)) \cdot 1499$
 $= 54628.00 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (17.2 + 0 + 1 + 169 + (18.2)) \cdot 1499$
 $= 308146.59 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 356 \cdot 10 \cdot 0.49 \cdot 1499$
 $= 41027. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 656 \cdot 44 \cdot 0.49 \cdot 1499$
 $= 332814. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 170) \cdot (16 - 0) \cdot 0.7 \cdot 1499$
 $= 89557. \text{ kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 356 \cdot 58 \cdot 0.74 \cdot 1499$
 $= 359365. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 356 \cdot (58 - 1) \cdot 0.74 \cdot 1499$
 $= 353170. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (332814 + 89557) = 422371 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (41027 + 359365 + 353170 + 0) = 753562 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (332814 + 353170 + 0) = 685984 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 422371 kgf, must exceed W = 224055 kgf or W1 = 280805 kgf
Path 2-2 = 753561 kgf, must exceed W = 224055 kgf or W2 = 54627 kgf
Path 3-3 = 685983 kgf, must exceed W = 224055 kgf or W3 = 308146 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6068 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 6.2767 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 564.2767 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 4.711 %

Input Echo, WRC107/537 Item 1, Description: C-14 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	4950.000	mm
Vessel Thickness	Tv	58.000	mm
Design Temperature	T1	430.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1498.98	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm
Nozzle Diameter	Dn	355.600	mm
Nozzle Thickness	Tn	16.000	mm
Nozzle Material		SA-387 11	
Nozzle UNS Number		K11789	
Nozzle Cold S.I. Allowable	SNmc	1509.10	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1498.98	kgf/cm ²
Thickness of Reinforcing Pad	Tpad	58.000	mm
Diameter of Reinforcing Pad	Dpad	655.600	mm
Design Internal Pressure	Dp	6069.965	kPa
Include Pressure Thrust		No	

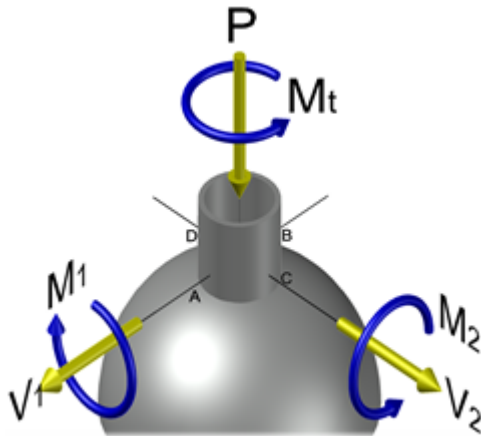
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	3211.0	kgf
Longitudinal Shear (SUS)	(V1) V1	2319.1	kgf
Circumferential Shear (SUS)	(Vc) V2	2319.1	kgf
Circumferential Moment (SUS)	(Mc) M1	2497.5	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	2497.5	kgf-m.
Torsional Moment (SUS)	Mt	2872.2	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(2504.5 (58.0 - 1.0))}$$

$$= 1602.443 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	3211.0	kgf
Circumferential Shear	(VC) V2	2319.1	kgf
Longitudinal Shear	(VL) V1	2319.1	kgf
Circumferential Moment	(MC) M1	2497.5	kgf-m.
Longitudinal Moment	(ML) M2	2497.5	kgf-m.
Torsional Moment	MT	2872.2	kgf-m.

Dimensionless Param: U = 0.33 TAU = 10.61 RHO = 4.00 (7.19)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.06038	(A,B,C,D)
$M(x) / P$	SP 4	0.02476	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.18034	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.08006	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.18034	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.08006	(A,B,C,D)
$N(y) * T / P$	SP 4	0.33991	(A,B,C,D)
$M(y) / P$	SP 4	0.12561	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.27645	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	0.46910	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.27645	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	0.46910	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5
Rad.	Bend. P	-3.6	3.6	-3.6	3.6	-3.6	3.6	-3.6	3.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-6.3	-6.3	6.3	6.3
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-16.8	16.8	16.8	-16.8
Rad.	Memb. ML	-6.3	-6.3	6.3	6.3	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-16.8	16.8	16.8	-16.8	0.0	0.0	0.0	0.0

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-7.7	-7.7	-7.7	-7.7	-7.7	-7.7	-7.7	-7.7
Rad.	Bend. P	-24.8	24.8	-24.8	24.8	-24.8	24.8	-24.8	24.8
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-18.7	-18.7	18.7	18.7
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-117.0	117.0	117.0	-117.0
Rad.	Memb. ML	-18.7	-18.7	18.7	18.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-117.0	117.0	117.0	-117.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-168.2	115.4	103.2	-81.3	-168.2	115.4	103.2	-81.3
Tang. Memb. P		-2.3	-2.3	-2.3	-2.3	-2.3	-2.3	-2.3	-2.3
Tang. Bend. P		-7.7	7.7	-7.7	7.7	-7.7	7.7	-7.7	7.7
Tang. Memb. MC		0.0	0.0	0.0	0.0	-5.6	-5.6	5.6	5.6
Tang. Bend. MC		0.0	0.0	0.0	0.0	-35.0	35.0	35.0	-35.0
Tang. Memb. ML		-5.6	-5.6	5.6	5.6	0.0	0.0	0.0	0.0
Tang. Bend. ML		-35.0	35.0	35.0	-35.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-50.6	34.8	30.5	-24.1	-50.6	34.8	30.5	-24.1
Shear VC		4.0	4.0	-4.0	-4.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-4.0	-4.0	4.0	4.0
Shear MT		7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5



Tot. Shear	11.4	11.4	3.5	3.5	3.5	3.5	11.4	11.4
Str. Int.	169.3	117.0	103.4	81.5	168.3	115.6	104.9	83.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		667.3	667.3	667.3	667.3	667.3	667.3	667.3	667.3
Rad. Pl (SUS)		-7.8	-7.8	4.8	4.8	-7.8	-7.8	4.8	4.8
Rad. Q (SUS)		-20.4	20.4	13.2	-13.2	-20.4	20.4	13.2	-13.2
Long. Pm (SUS)		667.3	667.3	667.3	667.3	667.3	667.3	667.3	667.3
Long. Pl (SUS)		-17.9	-17.9	1.4	1.4	-17.9	-17.9	1.4	1.4
Long. Q (SUS)		-116.8	116.8	80.2	-80.2	-116.8	116.8	80.2	-80.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		3.6	3.6	-3.6	-3.6	-3.6	-3.6	3.6	3.6
Shear Q (SUS)		12.6	12.6	12.6	12.6	12.6	12.6	12.6	12.6
Pm (SUS)		667.3	667.3	667.3	667.3	667.3	667.3	667.3	667.3
Pm+Pl (SUS)		660.7	660.7	674.4	674.4	660.7	660.7	674.4	674.4
Pm+Pl+Q (Total)		641.5	769.1	750.1	660.0	639.8	767.1	752.8	662.5

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	667.28	1498.98	Passed
Pm+Pl (SUS)	674.41	2248.47	Passed
Pm+Pl+Q (TOTAL)	769.07	4496.94	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1344.8	1344.8	1344.8	1344.8	1344.8	1344.8	1344.8	1344.8
Rad. Pl (SUS)		-26.4	-26.4	11.0	11.0	-26.4	-26.4	11.0	11.0
Rad. Q (SUS)		-141.8	141.8	92.2	-92.2	-141.8	141.8	92.2	-92.2
Long. Pm (SUS)		1344.8	1344.8	1344.8	1344.8	1344.8	1344.8	1344.8	1344.8
Long. Pl (SUS)		-7.9	-7.9	3.2	3.2	-7.9	-7.9	3.2	3.2
Long. Q (SUS)		-42.7	42.7	27.3	-27.3	-42.7	42.7	27.3	-27.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.0	4.0	-4.0	-4.0	-4.0	-4.0	4.0	4.0
Shear Q (SUS)		7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5
Pm (SUS)		1344.8	1344.8	1344.8	1344.8	1344.8	1344.8	1344.8	1344.8
Pm+Pl (SUS)		1337.8	1337.8	1357.5	1357.5	1337.8	1337.8	1357.5	1357.5
Pm+Pl+Q (Total)		1295.3	1461.9	1448.2	1321.0	1294.3	1460.4	1449.8	1323.0



Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1344.85	1498.98	Passed
Pm+Pl (SUS)	1357.46	2248.47	Passed
Pm+Pl+Q (TOTAL)	1461.87	4496.94	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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